

Enceladus Mission - PDR

NASA L'SPACE ACADEMY
Team 20 - 'Oumuamua'

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Table of Contents

Introduction & Summary

1.1. Team Introduction.....	5
1.2. Mission Overview.....	7
1.2.1. Mission Statement.....	7
1.2.2. Mission Requirements.....	7
1.2.3. Mission Success Criteria.....	8
1.2.4. Concept of Operations (COO).....	9
1.2.5. Major Milestones Schedule.....	9
1.3. Descent Maneuver & Lander Summary	11
1.4. Payload & Science Instrumentation Summary.....	13

Evolution of Project

2.1. Evolution of Descent Maneuver & Lander.....	15
2.2. Evolution of Payload & Science Instrumentation.....	16
2.3 Evolution of Mission Experiment Plan.....	17

Descent Maneuver & Lander Design

3.1. Selection, Design, & Verification.....	18
3.1.1. System Overview.....	18
3.1.2. Subsystem Overview.....	19
3.1.3. Dimensioned CAD Drawing of Entire Assembly.....	20
3.1.4. Manufacturing & Integration Plans.....	21
3.1.5. Verification & Validation Plans.....	23
3.1.6. FMEA & Risk Mitigation.....	25
3.1.7. Performance Characteristics & Predictions.....	29
3.1.8. Confidence & Maturity of Design.....	29

3.2. Recovery/Redundancy System.....	31
3.3. Payload Integration.....	33

Payload Design & Science Instrumentation

4.1. Selection, Design, & Verification.....	34
4.1.1. System Overview.....	34
4.1.2. Subsystem Overview.....	34
4.1.3. Manufacturing Plan.....	36
4.1.4. Verification & Validation Plan.....	37
4.1.5. FMEA & Risk Mitigation.....	38
4.1.6. Performance Characteristics.....	40
4.2. Science Value.....	41
4.2.1. Science Payload Objectives.....	41
4.2.2. Creativity/Originality & Significance.....	41
4.2.3. Payload Success Criteria.....	43
4.2.4. Experimental Logic, Approach, & Method of Investigation.....	44
4.2.5. Testing & Calibration Measurements.....	45
4.2.6. Precision of Instrumentation, Repeatability of Measurement, & Recovery System.....	46
4.2.7. Expected Data & Analysis.....	49

Safety

5.1. Personnel Safety.....	51
5.1.1. Safety Officer.....	51
5.1.2. List of Personnel Hazards.....	51
5.1.3. Hazard Mitigation.....	52
5.2 Lander/Payload Safety.....	54
5.2.1. Environmental Hazards.....	54
5.2.2. Hazard Mitigation.....	55

Activity Plan

6.1 Budget.....	57
6.2 Schedule.....	59
6.3 Outreach Summary.....	60
6.4 Program Management Approach.....	61
Conclusion.....	63

1. Introduction and Summary

1.1 Team Introduction

Mattison Boveri: *University of Pennsylvania, Philadelphia, Pennsylvania*. Mattison is a sophomore studying Chemical and Biomolecular Engineering. She is also passionate about energy and sustainability and would like to pursue a career in the unique intersection of sustainability and space exploration. She is a current member of the Penn Aerospace Club in the balloon sector and will bring her chemistry and problem-solving knowledge to the team. Throughout the duration of this project, Mattison hopes to learn from the concept mission and her fellow teammates, who each bring something unique to the group.

Yuri Kim: *University of Massachusetts Amherst, Amherst, Massachusetts*. Yuri is a freshman studying Computer Science. Her excitement for technology and ideation first sparked in high school, after taking her first computer class. Equipped with a strong foundation in mathematics, computer programming, and design through summer classes, hackathons, and programs like Google CSSI, Yuri aims to apply her technical and teamwork experience to delve into this concept mission and learn as much as possible.

Kevin Ntoni: *Middlebury College, Middlebury, Vermont*. Kevin is a freshman studying computer science and economics. He has always been mathematically inclined and participated in robotics all throughout highschool. Kevin hopes to learn a lot and let his high energy and passion for space be a catalyst to his teammates.

German Paez Garcia: *The Ohio State University, Columbus, Ohio*. German is a senior Aeronautical and Astronautical Engineering student. Over his career, he has interned at AERTEC Solutions, been a manufacturing specialist at the Buckeye Space Launch Initiative, worked in the aerodynamics department of the Buckeye Current Racing Team, and currently holds a position in SGT's Aerospace Honorary Society Executive Board (OSU Chapter). Thanks to all the experience gained, German aims to apply all this knowledge into the design of this concept mission. As well as being able to learn as much as possible throughout the duration of the project.

Julian Perez-Doval: *Dartmouth College, Hanover, New Hampshire*. Julian is a sophomore studying Engineering Sciences and Economics. Driven to make a strong impact through the use of science, math, and analytics, he recently became interested in the space industry for his future goals and endeavors. Julian has teamwork and analytic experience through his involvement in trading and consulting clubs on campus, and he has developed technical/science writing skills through his first-year college writing seminar "Life on Mars".

Abigail Roossien: *Calvin University, Grand Rapids, Michigan*. Abigail is a sophomore studying Mathematics and Global Development. She has always had a passion for learning about anything space-related, but just recently decided to pursue this more as a future career. With a strong foundation in Math and a passion for learning new skills, Abigail hopes to learn a lot through this program.

Mirelys Rosales: *University of Maryland, College Park, Maryland*. Mirelys is a freshman studying Astronomy and Physics. She has a strong background in research and technical writing, developed from her participation in programs like the Ulysses Signature Program, as well as connections to several current researchers. Though this is her first big step into her desired field of study, she adapts quickly to the virtual environment provided and is eager to contribute all she can.

Peter Stokes: *University of Connecticut, Storrs, Connecticut*. Peter is a junior studying mechanical engineering. His love of space and space technology was sparked by an internship with NASA Langley's ACS3 team, and here he learned about the process NASA teams go through to make a mission come to fruition. Peter also interned at KT Corporation where he did both chemical and

mechanical engineering for projects centered around water filtration and learned technical writing skills as well as how to apply engineering to accomplish the goals of the project.

Eduardo Vargas Gutierrez: *Tufts University, Medford, Massachusetts*. Eduardo is a junior studying Mechanical Engineering and Mathematics. Eduardo has increasingly become interested in space exploration, specifically in making space travel faster, cheaper and overall more efficient. Eduardo has developed a great sense of collaboration, leadership and innovation through his internship at Northrop Grumman Space Systems and from working for 2 years in a research lab. Through this program, Eduardo aims to further his interests in mission concepts and protocols to learn new skills and insights which he can hopefully one day apply in future missions.

Luay Wadie: *Oakland University, Rochester, Michigan*. Luay is a senior majoring in Computer Science with an academic focus in the Computational Intelligence track. His skill-based development has been driven by his undergraduate research in artificial intelligence and machine learning. Luay is currently a NASA intern with projects relating to deep learning and software development. His skill set ranges from data exploration, analysis, and applications to detect anomalies and patterns within unexplored sectors. Through this program, Luay will aim to gain a greater understanding of how missions are prepared, investigated, and applied throughout the entire process from development to deployment.

Aina Zaidi: *University of Michigan, Ann Arbor, Michigan*. Aina is a sophomore studying Computer Science. She became interested in robotics and software after participating in a high school robotics team that sparked an interest. She is experienced with technical writing, a skill she will utilize in the documentation of the PDR. Through experience gained from past projects and research endeavors, Aina aims to further her knowledge of computer science as well as venture into uncharted territory to gain new experiences and skills.

1.2 Mission Overview

1.2.1 Mission Statement

In addressing the needs and specifications for the Enceladus Strategic Science Investigation mission, the team first developed the framework for the expedition. This included determining the data of interest and goals for the mission that will help humans better understand the solar system. The main objective is to search for different biosignatures and organics that can survive in Enceladus' environment, enabling us to find any evidence of life. To make the mission more attractive, the team will also analyze carbon dioxide, ammonia, and the temperature of the water vapor and ice particles spewed through Enceladus' plumes. The clues provided by this information may inform us whether the core's composition is similar to that of a carbonaceous chondrite, thereby indicating whether this heat source originates from a fairly recent heating event or from the beginning of Enceladus' life. With the data to be sampled, scientists also hope to determine the source from which this heat originates.

1.2.2 Mission Requirements

1. Launch Constraints

- R1. The payload shall have a mass under 77kg at launch.
- R2. The payload shall fit within a 51cm x 51cm x 76cm volume at launch.
- R3. The mission shall cost no more than \$400 million.

2. Landing

- R4. The lander shall perform a deorbit burn to target a predetermined landing site.
- R5. The lander shall propulsively land so the force of impact creates no more than 1 g of acceleration.
- R6. The lander shall land leveled on its landing legs.
 - a. The lander's landing gear shall be lowered 2-3 minutes before landing.
- R7. The landing site shall be within 5 km of the target location.

3. Science

- R8. The mass spectrometer shall analyze at least 5 samples and record the results.
- R9. The cameras shall take and transmit pictures of the landing as well as the surface after landing.
- R10. The radio transmitter shall transmit the data from all instruments and sensors to Earth within a day of the measurements being taken.

4. Engineering

- R11. The lander shall keep instruments and its "vital organs" within their operating temperatures (-40° C to 40° C).
- R12. The lander shall take and transmit the results of measurements for a week before running out of power.
- R13. The lander shall be able to resist damage from water and dust found on the surface.
- R14. The lander shall contain a rechargeable power source or a fixed power source with enough energy to carry out the desired science.
- R15. The lander's power source shall output peak power of at least 25 watts and a total capacity of 10 amp-hours.

1.2.3 Mission Success Criteria

When considering the criteria for success, the team took into account the main goals (outlined in the Mission Statement) for this mission. The most important task is to sample data from at least five different locations within the Tiger Stripe region in the southern pole. Results received from these five data samples will be investigated and compared to establish reliability; more specifically, a biosignature will be successfully identified if it is recorded in at least two spots.

Landing in a new body within the solar system presents unforeseen challenges, and more so with an ambitious plan to access five different areas within the Tiger Stripe region. Therefore, the instrumentation/machines dispatched should return data back to Earth from at least four of the five locations. This criterion ensures a reliable comparison of biosignature and organic readings between our mission and the Cassini mission.

To test the idea of heating through rock-water interactions, at least two different spots should communicate readings of silica particles between the currently known limited size range of two to eight nanometers in radius. Silica particles with radii outside of this range should also be considered and monitored.

Another failure that we have perceived as a critical requirement is the ability to safely and reliably transmit the gathered digital data back to the orbital satellite to then be transmitted back to Earth. The entire mission revolves around this phase to occur without any technical issues as the core dependency is to investigate the surface of Enceladus as well as the vast oceanic universe below its frozen layers.

Any obstacle that severely hampers or completely eliminates the mobility of the lander will be considered a failure. An integral part of this mission is the concept of gathering data from different spots within the Tiger Stripe Region to build reliability, consistency, and opportunity. The less spots that are inspected, the higher the probability that results may be less reliable and the lower chance for running into accidental discoveries that may inform our mission.

The final failure criteria identified is for the lander to keep operational temperature conditions for the instrumentation once it lands at Enceladus' surface. The whole concept of the mission is to sample and analyze data which will help us identify any kind of micro-organism life on this moon. If the scientific instrumentation is not kept at an appropriate operational temperature, it will fail to function properly due to the extreme freezing conditions faced in Enceladus. Therefore, it will be considered a failure if there is a malfunction of the system and the instrumentation temperature conditions cannot be sustained.

1.2.4 Concept of Operations

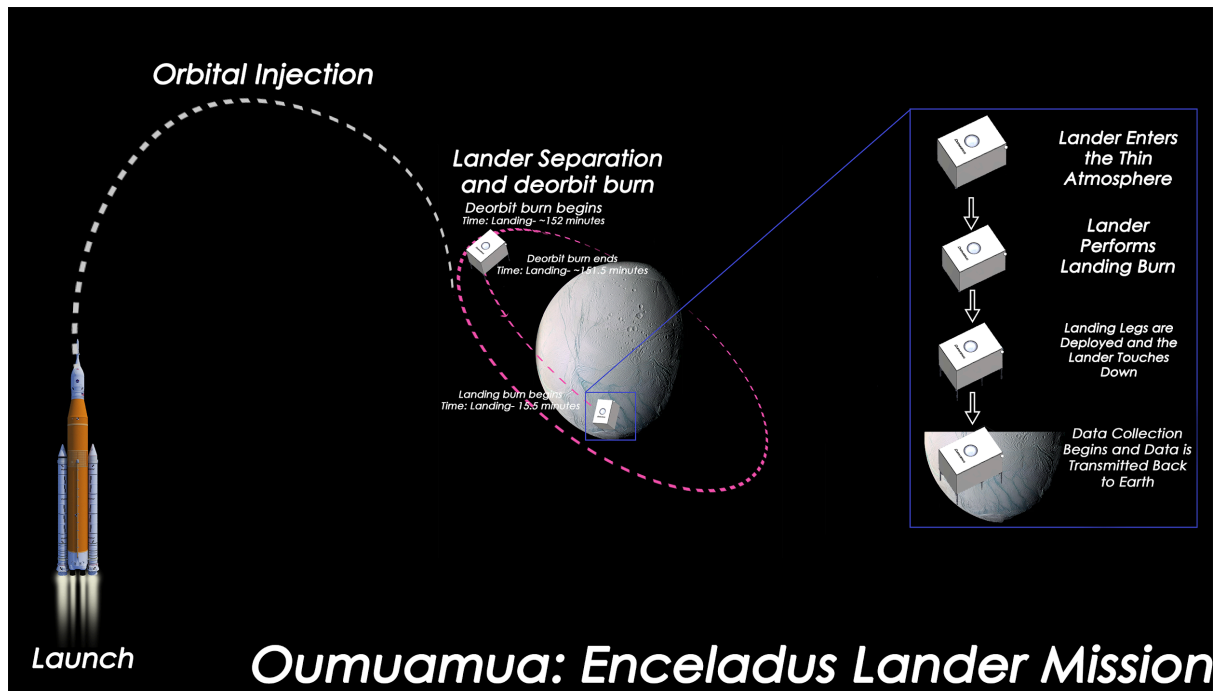


Figure 1: Concept of Operations

Sources: <https://www.jpl.nasa.gov/spaceimages/details.php?id=PIA06254>
<https://www.boeing.com/thefuture/space/index.html>

Figure 1 shows the Concept of Operations for the Lander. It begins as a secondary payload that is eventually carried into an elliptical polar orbit around Enceladus. From this position, the lander separates from the satellite and performs its deorbit burn. After the burn, the Lander enters the thin and watery atmosphere of Enceladus and will perform its landing burn to scrub off the rest of its velocity. The legs are then deployed to assure a safe landing into Enceladus' surface, where it will then begin the collection of data.

1.2.5 Major Milestones Schedule

The mission to Enceladus will be one that requires much thought, research, and oversight. The process will be several months long and require multiple phases:

- **Pre-Phase A, Conceptual Study:** Throughout this phase, the team will study Enceladus with detailed lenses. The weekly L'Space sessions will provide some background to the subject but outside research also has to be conducted. This phase starts on September 1st, 2020 and is expected to end on December 3rd, 2020. The summary and evolution of the project will also be detailed during this period.
- **Phase A, Preliminary Analysis:** From January 10th, 2021 to April 1st, 2021, the project will undergo the preliminary analysis. The first drafts of the project will be completed and sent to L'Space, as well as conducting more research reviews, to further develop the team's mission concept. Taking into the given design limitations, the team will research and plan out a detailed scope of their proposed spacecraft, analysis tools and its proposed route to Enceladus. The team will also locate areas of testing that will take place in Phase B in order to ensure a safe and successful mission. By the end of Phase A, the team will have a solidified proposal ready for outside review, testing, and development. After Phase A, the proposal will go under NASA review until approximately June 2021 before moving to Phase B.

- **Phase B, Definition:** From June 8th, 2021 to September 18th, 2021, if the team's proposal is taken under consideration, the team will undergo the System Requirements Review, System Design Review and Non-Advocate Review, focusing on the items of interest specified in Phase A. New experiment criteria is also taken under consideration from team members. The proposal will be sent back to NASA for peer group review, along with a final review of the experiment selection from NASA Headquarters' Science Mission Directorate (SMD) Steering Committee (SC). Instruments will then be created and evaluated by their designated teams.
- **Phase C/D, Design and Development:** From August 31st, 2021 till launch day (August 20th, 2022), the building and development of the spacecraft takes place. Using the ATLO (Assembly, Test, and Launch Operations) process, the spacecraft will be constructed before undergoing testing, launching, deployment, and verification. Like other spacecraft, it will also undergo a simulated space environment prior to launch. These tests will ensure high performance and a successful mission. Furthermore, all analysis tools and ground systems will be constructed and implemented to test alongside the spacecraft.
- **Phase E, Operations Phase:** Phase E, also known as the Mission Operations and Data Analysis phase, takes place from the launch of the spacecraft until its return. This phase lasts from September 30th, 2022 till April 16th, 2023. The phase consists of flying the spacecraft and guiding it on its planned course along with obtaining the scientific data laid out in Phase A. This critical phase takes the bulk of time and will extend from launch till return.

1.3 Descent Maneuver and Lander Summary

In order to successfully conduct a descent for the spacecraft, the team had to research various factors such as the atmospheric pressure and landing procedures. While the initial plan was to implement a parachute landing, it was later determined that there is not sufficient information about Enceladus' atmosphere to conduct an effective parachute landing. While research is still underway regarding the atmospheric pressure, it is still a factor that is largely unknown about and is one of the topics of interest for when the spacecraft arrives on Enceladus. Due to this issue, the team decided to conduct a propulsive soft landing instead.

The propulsive landing will be regulated mainly through the use of thrusters to accomplish a soft landing. In order to operate the thrusters successfully, the team will need to control weight, thrust, and throttle. In addition to these factors, the team will have to account for the climate and composition of Enceladus to ensure optimal landing procedures. Due to the icy nature of Enceladus' southern polar region, deorbit burn will be performed by firing the thrusters retrograde at apoapsis to make the periapsis intersect with the desired landing site. This procedure will focus on guiding the lander descent to avoid icy boulders, ongoing plumes, or other potential hazards. Additionally, shortly before impact, thrusters will be enabled in two directions: retrograde, as well as the opposing direction from the surface of Enceladus. This process will serve to eliminate tangential velocity in order for the spacecraft to land upright, as well as decrease the speed of the vertical velocity so that the lander is capable of landing level and absorbing the remaining velocity in its landing legs.

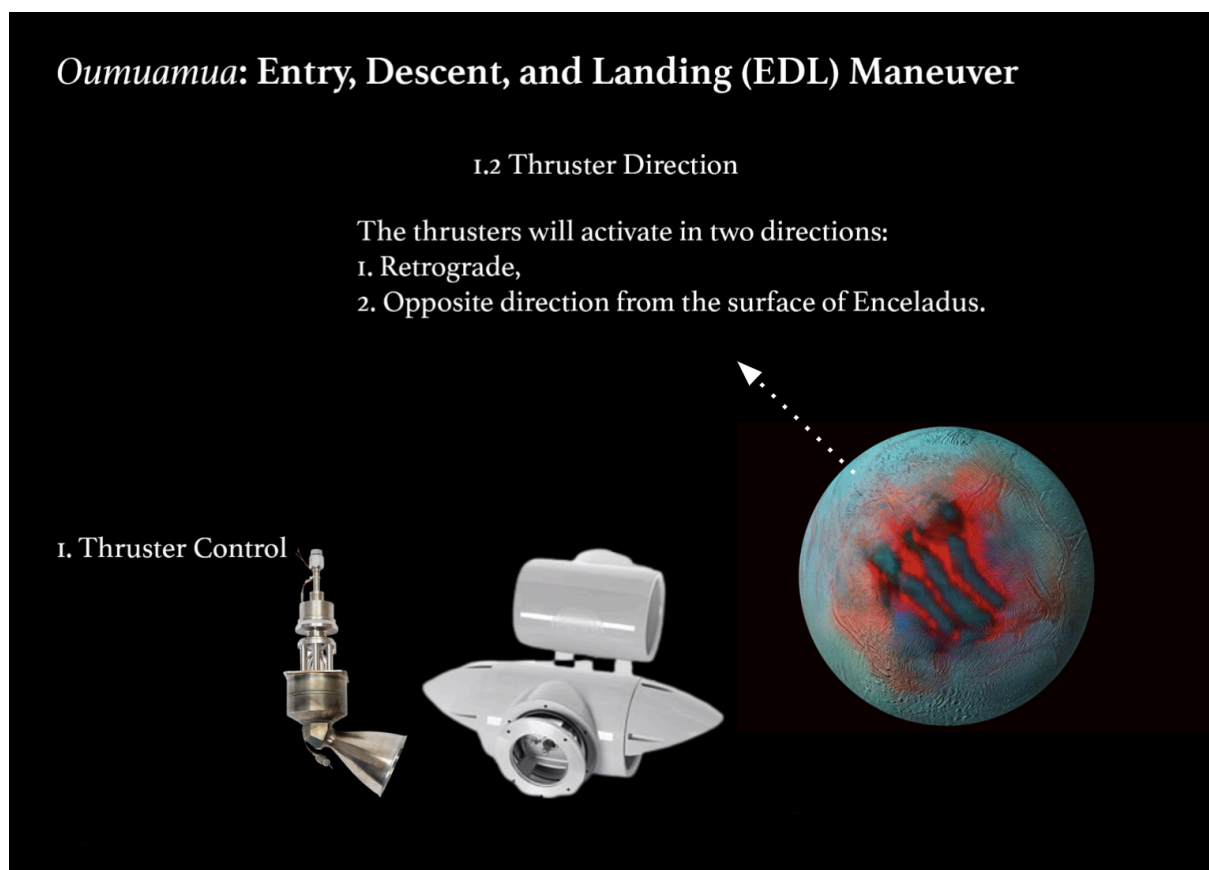


Figure 2: EDL Maneuver

Sources:

NASA/JPL-Caltech/University of Arizona/LPG/CNRS/University of Nantes/Space Science Institute
<https://www.space-propulsion.com/spacecraft-propulsion/hydrazine-thrusters/index.html>
https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.hiclipart.com%2Fsearch%3Fclipart%3Dthruster&psig=AOvVaw2Yhu72mGRzd8sD_hCrTFi6&ust=1602627208600000&source=images&cd=vfe&ved=0CAMQjB1qFwoTCKCl6d6JsOwCFQAAAAAdAAAAABA

Due to the time constraints of the project, the current lander's mass is required to be no more than 77kg, though some mass will be lost with the propellant used for landing. Using $g = GM_E/R^2$, where M_E is the mass of Enceladus and R the radius, the gravitational acceleration at the surface was determined to be 0.11 m/s^2 . With an altitude of 35km, the vertical velocity component of the lander at the surface was found to be around 90 m/s. This, combined with the 177.2 m/s velocity at periapsis, would require a ΔV of around 200m/s. The ISP of most monopropellants is around 200s, so for a conservative estimate the lander is initially designed to have 17kg of fuel to ensure there is sufficient power to land safely [1]. This equates to over 400m/s of ΔV , meaning that the dry weight of the lander should have a limit of 60kg. The volume will be $197,676 \text{ cm}^3$, or no more than 51cm x 51cm x 76cm. The entry angle is estimated to be shallow at 15° , modeled off from Mars' *Pathfinder*, which utilized a small angle to avoid descent tasks lacking sufficient time to prepare a successful landing [2]. The lander will leave orbit with an entry speed of around 180 m/s and an altitude of 35 km. These factors will likely change based on the orbital path to Enceladus, a largely uncharted territory that will require further research before implementation.

1.4 Payload and Science Instrumentation Summary

In addressing the internal systems and science instrumentation needed for the mission, the team had to take into account the main goals for the expedition. Since the main purpose is to determine whether there is life in Enceladus or not, detecting biosignatures will be one of the focuses. To do so, the team outlined the importance of carrying a Gas Chromatography-Mass Spectrometry instrument in the Lander. This apparatus would process the sample by first sending it through a gas chromatograph that would define the sequence and rate of separation from the molecules, and would later enter it into the mass spectrometer. Here, the molecules will be evaluated and identified. Once the sampling is finished the data will be converted into a signal and transmitted to Earth. The GC-MS, equipped with a high-voltage quadrupole for its exceptionally high specificity detector, would also serve to analyze aerosols during descent through its filters and pyrolyzation, as well as to measure the composition of Enceladus through its heating before impact, which would subtly vaporize surrounding surface material upon landing to make the surface readable without causing mission-compromising effects to that surface. If a GC-MS similar to the Mars Organic Molecule Analyser (mass-spectrometer-based instrument) apparatus is used, we can be looking at a mass of 11.5kg. The downside would be that the operating temperature range is only from 233K-293K, much higher than the expected temperatures at Enceladus's surface. In order to account for this, the team will try and develop a heating mechanism to keep the GC-MS in operational temperature conditions. The average power use is of 65W and the maximum power output can reach up to 154 W. Finally, no volume parameters were found for that specific apparatus, but they were found for a similar one; the Ion and Neutral Mass Spectrometer from the Cassini-Huygens mission. The volume is estimated to be 0.0312m³, therefore making it a feasible instrument to carry in the Lander.

When assessing the temperatures from the water vapor and ice particles being spewed through Enceladus's plumes, an Infrared Spectrometer will be used. Such apparatus will provide color, composition and thermal maps on the surrounding environments from the Lander. This is done by recording the amount of energy as a function of the wavelength from the infrared radiation that passes through the sample. Therefore this Infrared Spectrometer will allow us to record the temperature of the molecules being spewed through Enceladus' plumes, fulfilling one of the mission objectives. A similar instrument, called Ralph, was used in the New Horizons probe launched in 2006. Such spectrometer weighed 10.5kg, had a maximum power use of 7.1 Watts and a volume of 0.0593m³. Taking into reference these specifications, the team should be able to apply such an instrument in our mission since it does not suppose a great use of space nor weight in the Lander's payload.

The power source that will accompany this mission to Enceladus will be a standalone lithium battery for the duration of the mission. These types of batteries were used on Sojourner rover on the Pathfinder mission which allowed it to recharge on solar arrays. For our case, we won't utilize charging on these battery cells but rather attempt and offset the decaying charge of these batteries by utilizing solar arrays while traveling through space to Enceladus. This battery will be able to carry 40-amp-hour, at 12V would result in 480 watt-hours to be utilized for the mission. This size comes at 5.9 kg, which in turn doesn't seem to be a lot and we can always increase the battery's amp-hour if we require more input by purchasing a higher capacity. As we explore our instrumental needs as well as the rover's critical life systems such as temperature regulation and experimentation assessments, we will attempt to utilize specific volume dimensions to fit our desired power source needs.

Cameras serve for navigation, hazard detection, as well as photographs. The lander will also have several sensors and cameras, including engineering, science and descent imaging cameras. Engineering cameras mounted to the rover's body will focus on hazard avoidance and navigation after landing. They will weigh 9 ounces each with an image size of about 1024x1024 pixels, covering red wavelengths centered at ~650 nanometers. The camera module is very small and only adds a volume of .0000368m³.

Now that all the main payload and science instrumentation to be used for the mission have been summarized, we can evaluate how they will fit in the missions lander. The sum of masses from the GC-MS, Infrared Spectrometer, SNAP-19 RGT and camera is estimated to be 40.6kg. Regarding

the volume that will be used, it will be expected to occupy approximately 0.116326m^3 . Since the maximum allowable mass and volume of the payload were determined to be 60kg (due to 17kg of fuel being consumed in the landing stage) and 0.1977m^3 respectively, the team will have more space to add other instrumentations in the Lander if needed. Since one of the goals is to collect samples from different locations, some of this “free” space we have could also be spent on more fuel or power generators to move the lander.

After reviewing all the critical instrumentation that will be used for the purpose of this mission, it now has to be determined how the samples are going to be collected and analyzed. The team opted to go with a robotic arm configuration that will extend from the main component of the lander. This arm will contain a small drill that will be used to make a small hole of several centimeters deep into the surface crust. The drill will be hollow, allowing for the surface particles to be collected whilst digging into the ground. Once the hollow space inside the drill has been filled up with sample particles, the robotic arm will insert the collected samples into the part of the lander where the scientific instrumentation will be stored.

2. Evolution of Project

2.1 Evolution of Descent Maneuver and Lander

The initial plan for the descent maneuver involved using a parachute to slow down. This was based on how the early and lighter Mars landers used parachutes and how this is a traditional method used on Earth to slow falling objects. Parachutes are relatively simple once deployed, as they will interact with the atmosphere without need for computers monitoring all details. This allows for a relatively passive descent except for deploying the parachute. Once the atmosphere of Enceladus was researched it became less appealing of an option. Enceladus releases vast amounts of water and some of it accumulates in a trace atmosphere near the tiger stripes, but this atmosphere is not thick enough to provide any significant drag. This would mean it is not an effective method on Enceladus, so a new design was created.

The next design focused on maximizing the controllability and reliability of the landing. This called for four monopropellant thrusters (one in each corner of the lander) each with their own fuel tank. These four thrusters could be throttled in order to change the attitude and steer the lander. Four thrusters would allow for some compensation if one was to fail. Each also was planned to have their own fuel tank in order to have some redundancy. This meant that even if one fuel tank lost pressure or otherwise failed to perform, three of the other thrusters could still fire and using clever throttling the impact on performance could be minimized. Four smaller fuel tanks also meant that the volume of the fuel tanks were spread out meaning it would be easier to fit them around the other components of the lander. As the fuel drained during flight, the center of mass would remain constant, as each fuel tank would be balanced by the opposite fuel tank. This helps increase simplicity and reduces the need to compensate for a changing center of mass.

Multiple fuel tanks was eventually changed in favor of a single, centralized fuel tank for a few reasons. The first reason is that it is far more efficient mass and total volume wise, as less material for the fuel tank container would be needed. It also reduces the total number of parts that could go wrong. A centralized tank also maintains a constant center of mass even during landing. This design helped free up mass and volume for other aspects of the lander.

2.2 Evolution of Payload and Science Instrumentation

The original power source that was going to accompany this mission to Enceladus was a Radioisotope Thermoelectric Generator (RTG). A RTG utilizes heat that is supplied from the decaying Plutonium which in turn transitions into Uranium and releases the alpha particles as the heat exchanger source between this interaction. But after several iterations on how to power the entire system, the team concluded the usage of batteries to power the mission objectives for several days to analyze and gather required mission requirements. The battery scope that we want to utilize within this mission could rely on previous like ones used by NASA missions. Lithium Ion batteries are powerful enough to sustain power decay over a long period of time during the duration to Enceladus, but we can always get more power to confiscate for the loss decay percentage upon arrival. We will use two batteries with 40-amp-hour at 12V which will result in 480 watt-hours for the mission requirements. This should suffice the requirements and can be scalable depending on the remaining requirements for the mission.

The whole landing mission concept has been subject to several modification iterations throughout the development of the project. Initially, it was stated that the lander would only sample data at one location. This was quickly changed at the beginning of the design process since the team wanted to make all the data obtained as reliable as possible. Therefore five landing spots were selected and a biosignature will be successfully identified if it is recorded in at least two spots. The final iteration made regarding the landing locations was made towards the end of the project. It was assessed the benefits and disadvantages of having the lander abandoned at the final location versus sending it to a “kamikaze” mission. This kamikaze mission would entail sending the lander into one of the plume sources in the Baghdad sulcus. Although it is almost guaranteed that the lander would be destroyed when impacting with the water flow being spewed through the plumes, it would be able to record further data and images from a closer perspective to this region.

At the start of the planning process, the plan was to gather data from the surrounding atmosphere of Enceladus. In order to avoid contamination, the plan was to keep distance from the surface of Enceladus itself. However, as the planning progressed, it was realized that in order for the quality and quantity of data to be optimal, the spacecraft would need to be in close contact with the source of the data. Therefore, it was decided that the spacecraft would land in the tiger stripes region on the south pole of Enceladus and a drill would be used to gather the necessary data. The implemented drill also for exact extraction of liquid, vapors, and solids without contamination of the environment of Enceladus. This will allow for high quality and quantity of information to be collected while the spacecraft is on Enceladus.

2.3 Evolution of Mission Experiment Plan

The science objectives from the team were also subject to change throughout the evolution of the mission plan. The main focus at the beginning of the project was to detect whether there are any biosignatures in Enceladus' surface, which would enable us to determine if there is some sort of life on this moon or not. It was later determined that we could set as a secondary goal to get a better insight into the heat source originated at Enceladus core. This would allow us to determine if these heat sources originate from a recent heating event or from the beginning of Enceladus' life. In case that the biosignatures proved to be inconsistent or ambiguous, these clues could potentially suggest if Enceladus harboured life at some point in the past.

The idea of having the lander visit different locations in the Tiger Stripe region was always present. Later on, it was decided that the lander would visit 5 different locations and recover information from each one of them. To do so it was originally suggested to use a rover type of design with wheels. But since there is not too much data gathered from Enceladus' surface and what it can be expected from there, the team opted to use a thruster-type configuration for the lander since there is a lot of extra fuel that would not be used during the initial landing. This would guarantee the lander to move to different locations as a rover system could face high adversity in an unknown ground.

The final iteration made was regarding how the sample data was going to be collected, since there could be a wide variety of options possible. Initially the team thought the easiest way to approach this would be by taking gas samples and recovering the ice particles as it fell from being spewed out from the plumes. When trying to see how all the instrumentation necessary for this type of data sampling would fit in the lander, the team realized it was not a feasible option since it did not fit inside the lander. It was then decided to use a vertical arm with a small drill in its tip, which would be located in the bottom section of the lander. The drill would then be able to collect some samples from Enceladus' surface. In this way, all the payload could be guaranteed to fit within the lander.

3. Descent, Maneuver, and Lander Design

3.1. Selection, Design, and Verification

Section 3.1.1. System Overview

The propulsive landing will be deorbited through the coordination of four total thrusters, one for each corner of the lander. The thrusters will be the MR-111G 4N (1.0 lbs) Rocket Engine Assembly from Aerojet Rocketdyne. This will allow for the lander to perform a soft landing on the surface of Enceladus. The initial deorbit burn will be a retrograde burn performed at apoapsis, and it will take around 20.9 seconds using the four 4N thrusters. This burn would place the periapsis of the orbit at the surface of Enceladus in the south pole near the desired landing site. Approximately 135 minutes later, there will be 17 minutes remaining until impact. The lander should be surveying the area and the computer should finalize calculations for the landing burn [1]. This burn should also be predominantly retrograde, as there will be little to no atmospheric drag to slow down the horizontal velocity. As horizontal velocity is scrubbed, the burn will turn to be more and more vertical, akin to a gravity turn a rocket performs to takeoff, but in reverse. This landing burn should take just over 15.5 minutes. This burn will reduce the velocity to about zero, and in the final stages of the burn, the computer will make adjustments to target the optimal landing location in the immediate vicinity and bring the lander for a soft touchdown. The total maneuver will use around 200m/s of delta-v or about 7kg of fuel. The landing legs will be deployed before landing, and they will each have the ability to buckle to absorb landing forces if the touchdown is rougher than anticipated. The thrusters should be shut down a few feet above the ground in order to prevent kicking up dust, water, or any other particles that may potentially harm the lander's instruments.

The fuel tank is spherical with an inner diameter of around 291mm in order to hold 15kg of hydrazine. 15% of the fuel tank volume must be used for a pressurant gas in order to maintain high enough pressure to the thrusters. The tank will be made from X2021 aluminum which requires a tank thickness of .065in, or 1.651mm. This aluminum would be designed to handle stress of 9150psi or $6.309 \times 10^7 \text{Pa}$ at a factor of the safety level of 4.04 [2]. This size fuel tank can provide enough fuel for a total of more than 450m/s of delta-v, meaning there is more than twice as much as needed to land, leaving plenty of fuel to hop to new locations for increased sample variety.

There is little to no atmosphere on Enceladus, so landing using drag to slow down for touchdown is not a viable option. This means that there are not many options for backup systems. The lander is designed with semi-redundant thrusters to help reduce the chance of total failure. Each thruster is capable of a maximum of 4N of thrust, for a total of 16N, which means that if one thruster fails, the remaining three can throttle appropriately and the computer can stabilize and accommodate. If two thrusters fail, the lander would eventually have a thrust-to-weight ratio of over 1 after enough propellant is burned. This means that if the failures happen close to landing and the remaining thrusters are diagonal from each other, the lander may still be able to land without crashing completely. In addition to having multiple thrusters, the lander will be equipped with landing legs that have the ability to compress under high loads. This will function similarly to the landing legs of the Apollo landers, which implemented honeycomb cartridges that could absorb some of the shocks of impact [3]. This type of landing system also has the advantage of being somewhat self-leveling, as legs that hit first will endure a larger portion of landing force, and compress more, accommodating for higher terrain. These legs will be designed to take as much load as possible within the weight and volume restraints in the hopes that even if the lander hits the ground at a higher than intended velocity, it would survive. The legs are designed out of aluminum in two main pieces. The outer section is a hollow square tube 1cm outer width with a wall thickness of .5cm and a total length of 45.5cm.

The rationale behind choosing a propulsive soft landing as the primary form of descent stems from the need for a system that will account for minimal atmospheric drag but unknown specific atmospheric conditions of Enceladus. Due to the absence of drag, thrusters will serve to slow the lander before the descent. The velocity at impact is thought to be too high for an airbag or similar

system to protect it if no thrusters are used. Additionally, propulsive landing components can be tested, controlled, and modified to improve chances of survival. Using a computer to monitor distance from the ground and properties of the terrain ensures that the thrusters can be throttled for a completely controlled landing. The sensors have to be designed and tested in a way that will model the terrain of Enceladus, with specific countermeasures to ensure that the temperature, texture, vibrations, atmospheric particles, or any other potential hazard cannot harm the lander or its instruments.

The surface of Enceladus is not mapped with great resolution. This, combined with the chance that the lander misses the targeted landing site, would require landing on uncharted terrain. Since the surface of Enceladus at the south pole is covered in non-uniform ice, many locations may be rough with large blocks of ice unsuitable for landing. A propulsive landing system allows for more control in the final selection of a landing location which can increase the probability of mission success.

There are no stages nor anything else that will detach from the lander. The thrusters will be built into the corners of the lander and the legs will unfold below the lander. All will be connected to the same main housing unit that will hold the scientific instruments and other components. The systems of the lander include the electrical systems that run the scientific instruments, that run the sensors and heaters to maintain the health of the lander, and the antenna.

3.1.2 Subsystem Overview

The four monopropellant hydrazine thrusters were chosen for their simplicity and reliability. The thrusters are only 1.48kg each and about 20cm long by 2cm wide [4]. A monopropellant is simpler than a bipropellant due to the fact that there only needs to be one propellant tank as well as the fact that it is not necessary to mix fuel and an oxidizer in a very specific ratio in order to ensure performance. The thrusters chosen are from the same line of thrusters that are used on the Curiosity sky crane, and while they are much smaller and less powerful than the ones used with Curiosity, many of the reasons they were chosen are the same [5]. These thrusters are purchased with many of the required subsystems to operate them, like the 28-volt dual seat solenoid valves. The thrusters also come with valve heaters and a catalyst bed heater that will all be secured into the electrical system of the lander. The thrusters are made to be self-contained, so they do not require additional intervention. They will be connected to a single shared fuel tank for increased simplicity [6].

$$A = W/\sigma$$

Equation 1: Area (A) needed to support a certain weight (W) on a material with a compressive strength σ .

Using Equation 1, compressive strength of 250MPa, and assuming each leg should be strong enough to hold the entire weight of the lander on Enceladus (8.7N including the weight of fuel), the area of the landing leg needs to be at least $3.5 \times 10^{-4} \text{ cm}^2$ [7]. The area of the proposed outer section is 0.19 cm^2 . The inner section is 0.89 cm and the outer section has a wall thickness of 0.195cm. This is a total area of $.54 \text{ cm}^2$ which is still much greater than the necessary thickness. Equation 2 addresses the buckling load of the leg.

$$WL^2/(\pi^2 E) = I$$

Equation 2: Second moment of inertia (I) required to support a weight (W) with a beam length (L) considering a modulus of elasticity, E.

Equation 2 shows the necessary second moment of inertia for the outer beam with a modulus of elasticity (69 GPa) and length (60 cm) is $2.6 \text{ E-}12 \text{ m}^4$, and the actual second moment of inertia is $2.9 \text{ E-}10 \text{ m}^4$ [8]. The inner beam has a second moment of inertia of $4.7 \text{ E-}10 \text{ m}^4$ which are both much greater than the necessary loads. Each leg is expected to only need to carry less than $\frac{1}{4}$ of the weight accounted for with peak loads being in a greater range, according to the calculations made. They will be more than strong enough, and the legs' internal systems will include a material that can deform to absorb some of the impact velocity.

Landing legs will be unfolded using 1 low power servo motor each. A locking mechanism consisting of a spring that will push an aluminum cylinder into a locking hole will prevent the legs from moving once in the deployed position. The landing legs will be stored in the folded position where they are flush along the long 76cm sides of the lander (two on each long side). They will unfold away from the body and subsequently be just under 76cm long. This configuration was chosen as it is the simplest design and is the longest legs that can be used with such a design. The simplicity helps prevent failures in deployment, and the length gives room for a crush core and helps protect the landing body from any hazards it may encounter.

3.1.3 Dimensioned CAD Drawing of Entire Assembly

The general structure of the lander is a rectangular prism that occupies most of the space made available by the volume constraints of the mission requirements. Pieces such as the landing legs and drill unfold or extend from their stowed position within the volume constraints in order to perform as they were designed. Figure 3 shows the positions of the legs and drill once at rest on the surface and taking samples. A uniform hull protects and insulates the internal components of the lander.

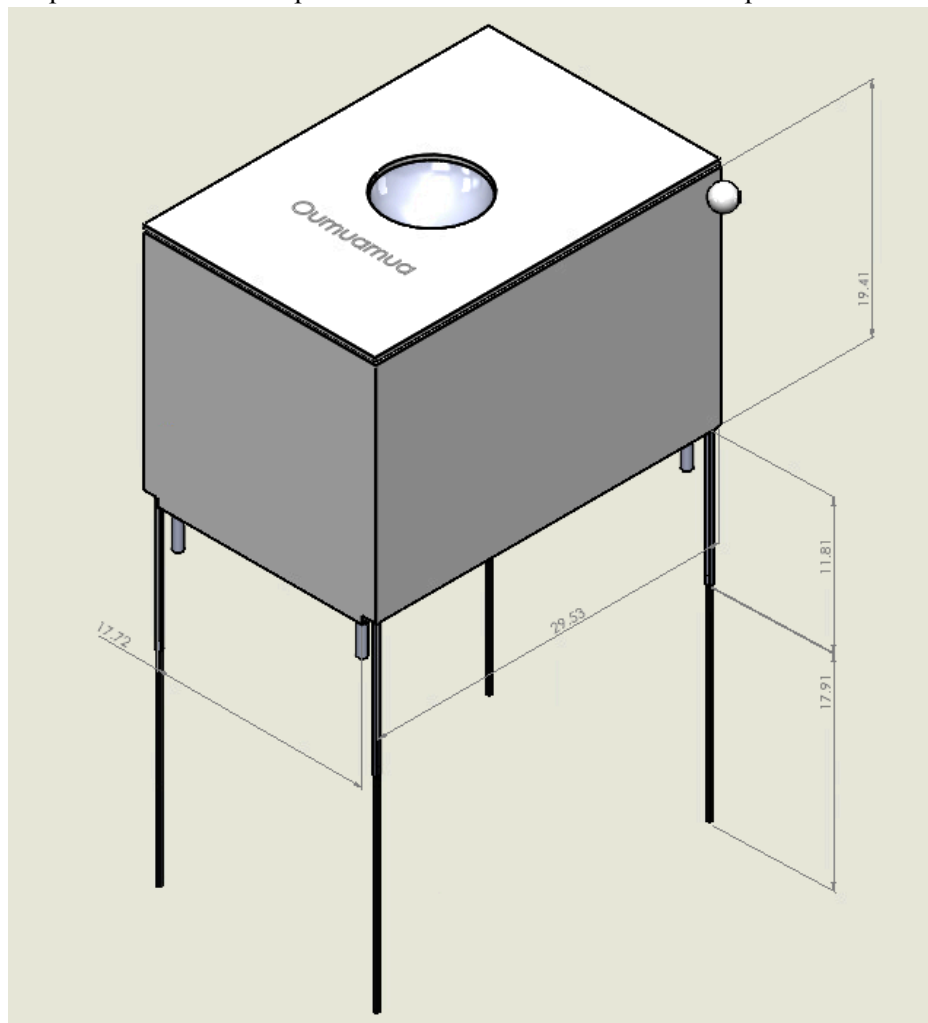


Figure 3: Lander with landing legs and drill extended. Shows dimensions in inches.

Figure 4 shows an exploded view of the lander. It fits just in the volume constraint and all components fit inside the hull snugly.

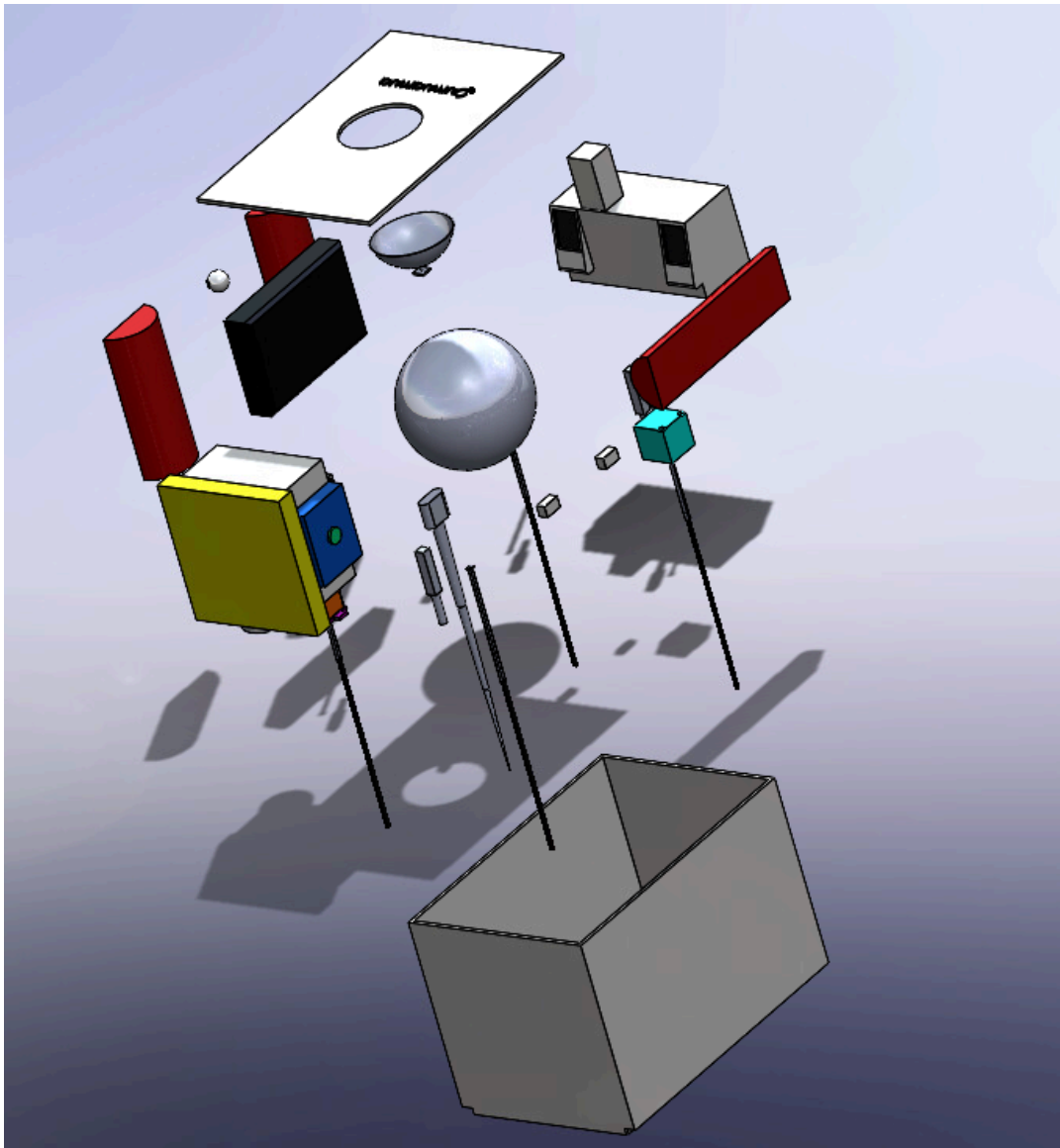


Figure 4: Exploded view of the lander and all components.

3.1.4 Manufacturing and Integration Plans

Taking place in Phase C/D, Design and Development, all manufacturing and integration plans will start from August 31st, 2021, until launch day (August, 20th, 2022). The first half will consist of final design planning and system requirement tests to prepare for a seamless system assembly, testing, and launch. Starting February 2nd, 2022, the team will begin manufacturing and integration plans. The lander's main body, including legs and other hardware, will be manufactured by the team's scientists and engineers, including NASA university-level co-ops and interns. Components of the computer, energy sources (including fuel tanks for thrusters), antennas for communication, and temperature controls, etc. will be built, manufactured, and assembled in a sterile field and, for some, in a vacuum-sealed environment [9]. Assembly of some components will be delayed while other instruments are assembled as manpower is limited. An efficient alternative is to outsource work for small and trivial parts to a manufacturing company, allowing the team to focus on the more complex parts of the lander.

All four Monopropellant Hydrazine thrusters will be manufactured by Aerojet Rocketdyne in a series production line. The thrusters will be on the lower end of thrust levels, possibly 4N or higher, due to

the low gravity on Enceladus [10]. Their combustion chambers and nozzles will be manufactured from Haynes 25 alloy to serve as a heat shield for protecting the propellant valve and lander. The thrusters will be equipped with an internally redundant catalyst bed heater and thermal insulation to guarantee optimum reproducible start-up conditions. The nominal valve voltage is 28 Volts DC. Their valves are dual seat solenoid type, produced by ArianeGroup and external partners.

The computer for the lander will interface with engineering functions of the lander over two networks that follow an aerospace industry standard designed especially for the high-reliability requirements of airplanes and spacecraft. The computer will contain a radiation-hardened central processor with PowerPC 750 Architecture to operate at up to 200 megahertz speed [11]. The computer will also contain 2.5 gigabytes of flash memory, 275 megabytes of dynamic random access memory, and 265 kilobytes of electrically erasable programmable read-only memory. The computer will constantly register the health of the lander including temperature and power levels to adjust temperature control accordingly and record power storage data. These parts will be manufactured by the team's scientists and engineers starting November 1, 2028, with certain smaller and more trivial parts being manufactured by other companies that specialize in computer manufacturing, such as Foxconn.

The lander will have low-gain and high-gain omnidirectional antennas for communication [12]. These will transmit waves at either low or high rate back to the antennas on Earth, allowing communication with the rover. Manufacturing of antennas will be conducted after assembly of the computer by the team's scientists and engineers for easy testing. However, if manufacturing of the computer is delayed, antennas will be manufactured by Antenna Products Corporation.

The lander's body or warm electronics box will consist of a strong, outer layer to protect the organs including batteries, electronics, the computer, and energy sources [13]. It will also keep the rover at a functioning temperature throughout the journey. This will ensure the lander does not "freeze to death." The lander will have radiators to release excess heat when in danger of overheating. It will be a heavy structure, weighing ~2,000 pounds. The body will be manufactured by the team's engineers and scientists, with the help of thermal fluids experts in engines. The manufacturing of the body will start once the manufacturing of the computer and antennas have commenced.

The lander will also have several sensors and cameras, including engineering, science, and descent imaging cameras [14]. Engineering cameras mounted to the rover's body will focus on hazard avoidance and navigation after landing. They will weigh 9 ounces each with an image size of about 1024x1024 pixels, covering red wavelengths centered at ~650 nanometers. The science cameras mounted to the rover's body will take color images and video footage, firing lasers and analyzing the elemental composition of vaporized minerals through the camera, providing close-ups of mineral structure and textures, and spectrometer. The descent imaging camera, attached to the lander, will detect the movement of the spacecraft and adjust it using thrusters for a safe descent. This camera will also sample the horizontal wind profile and create detailed geologic, geomorphic, and traverse planning and relief maps of the landing site. All cameras will be manufactured by the scientists and engineers from the team after manufacturing of the lander's body commences. The manufacturing of cameras will consist of machining and molding parts and PCB assemblies, followed by the assembly, testing, and calibration or alignments.

To account for extra time in the manufacturing process for testing of the lander, all manufacturing of individual parts will finish. Then, Phase D: System Assembly, Integration, and Test will launch. During this phase, the lander will go through tests including a simulated Enceladus environment. Some of the components of the lander, like the landing legs which will undergo buckling, will undergo multiple tests to continuously improve their performance. Specifically for the legs, the team will test for buckling failure through interactions with various materials at different temperatures and varying terrain. The lander legs will also include low power servo motors which will be acquired from a motor manufacturing company. The lander will undergo final testing that will be performed under an enclosed and unmanned environment involving wind, temperature, radiation, and several other simulations.

To ensure seamless integration of parts, the team will assemble components of exact dimensions from Siemens NX model. In addition to using modeling software, running force and pressure simulations will allow a greater understanding of what changes the team should implement. Through CAD and FEA simulation results, the team will initially build a smaller prototype lander of consistent scale before building the full-sized lander. This smaller model will also undergo scaled testing for performance assessment. Due to cost constraints, the team must ensure the Siemens NX model and the smaller scaled model provide us enough confidence to build the full-sized lander. Therefore, after extensive testing using CAD and the scaled lander, the team will develop the full-sized lander.

The estimated time for manufacturing and testing of the lander is nearly 7 months, with a margin to account for delays. This will account for the initial portion in which we build and test the system in Siemens NX, then build and test the scaled model, and then finally build and test the full-sized lander. Unexpected results may require additional manufacturing of parts and further extensive testing which is taken into account in the extra margin. On August 20th, 2022, all manufacturing and integration will have commenced and our lander will launch. This will allow us to launch by the planned date when Earth is at the most effective location relative to Enceladus.

Regarding the resources and man-hours needed, the budget has already accounted for a rough estimate for the cost of renting a manufacturing facility. For the man-hours needed, the main assumption is that the mission will be led by the 11 professionals writing this proposal plus 50 non-government workers who help with the manufacturing and logistics (since this is a relatively smaller mission).

3.1.5 Verification and Validation Plans

The team's validation and verification plans will be to simulate the lander's destination and new environment, as well as to test its overall performance. This will ensure that the lander was built correctly and that it is in fact, the appropriate lander for this mission. For this reason, the plan is to perform extensive testing in many conditions that can range from temperature testing to vibration and other part performance testing. The main goal is to simulate the harshest and possible conditions and challenges that the lander will face in Enceladus and ensure that it is able to withstand and overcome these.

The lander shall undergo "shake and bake" testing, which is meant to mimic the vigorous movements and conditions faced when entering Enceladus' atmosphere. For this type of testing, the lander will be placed on a vibration table and shaken intensely.

The lander will also undergo extreme temperature testing for environmental testing [15]. To be able to mimic conditions in space, the lander will undergo solar thermal vacuum testing where it will be exposed to extremely low and extremely high temperatures but mainly focused on a low temperature as Enceladus is extremely cold. The testing for low temperature will be at least -57 K and up to -127 K to ensure the lander is able to keep its body warm and functioning.

Another key test that the lander will need to complete is the testing of its telecommunication equipment for electromagnetic compatibility. It is of utmost importance that the lander's instruments can freely transmit and receive data without any interference.

The lander's ability to drill and detect life will also undergo extensive testing [16]. The drill shall be tested using different drill bits on different surfaces, as well as its ability to reach a goal depth. The mechanism for the lander's ability to gather, store, and analyze a sample shall also be tested for accuracy and timely delivery. For this testing, the team has planned to recreate the surface of Enceladus and attempt to have the lander drill through it. In addition, the Signs of Life detector shall also be tested for its ability to detect and identify the correct elements and samples of interest.

The outer “shield” or protective covering of the lander will undergo impact testing. This will involve tiny and relatively big objects to be launched at several different speeds at the lander’s outer shield. This testing will ensure the resiliency, hardness, and strength of the outer shield and its ability to survive and protect the entire lander from space debris and other potential collisions.

The legs of the lander shall also undergo landing testing to ensure that they can optimally absorb the impact from the landing. For this testing, the team will test different materials with different dimensions (length and thickness) and absorb how they buckle and perform when colliding into a surface at a specific speed while carrying a load similar (or exactly rather) to that of the lander body.

3.1.6 FMEA and Risk Mitigation

FMEA Chart: [FMEA Chart Link](#)

Functions	Failure Mode(s)	Effects	SEV	Cause(s)	Occ	Design Controls (Prevention)	Design Controls (Detection)	Det	RPN	Recommended Action(s)
To use four monopropellant hydrazine thrusters for a successful deorbit and smooth landing	One thruster or more fails to work during deorbit or landing	If one thruster breaks down, smooth landing is not possible but still likely. If two thrusters break down, smooth landing still possible if working thrusters are diagonal to each other. If three or more thrusters break down, possibility of crashing is extremely high.	8	Amount of "roughness" produced by ends in the catalyst bed may exceed the criterion of the mission, effectively shortening the life of the thruster	2	Same design as previous successful shuttle and satellite missions	Contact company to discuss fuel and thruster performance	6	96	
				Poisoning of the catalyst bed due to contact with a foreign substance	3	Power off thrusters a safe distance from the ground	Simulation test of landing	6	144	
To effectively control a computer system that is designed to make last minute adjustments for an optimal landing spot in the vicinity of the lander, including reliable sensor information retrieval	Computer system malfunctions	Computer system is unable to rewire landing spot as the lander approaches the surface of Enceladus, which may cause the lander to experience some damage upon landing	8	Computer system damaged if it is hit too hard by any debris	2	Construction of rover body to shield computer system from debris	Simulation tests for body shield impact resistance	7	112	
	Connectivity issues prevent communication lines between computer system and tech control group	Tech control group is not able to orchestrate where the lander will settle. In the case that the lander is on a trajectory to land on undesirable terrain, lander could be damaged	8	Bad signal (either because of distance, interference, etc.) between tech control group and computer system on lander	3	Use of low gain and high gain antennas to directly communicate from lander to the Deep Space Network (DSN)	Communications test between antennas and DSN	7	168	
	Sensors do not report information regarding temperature, texture, vibrations, atmospheric particles, or information is inaccurate	Even if tech control group can monitor where the lander will settle, inaccurate readings from sensors about the environment and nearby terrain can be an obstacle to landing in an optimal site	7	Sensors are damaged due to collision with debris or other material jumping up from the surface of Enceladus	3	Construction of rover body to shield sensors from debris	Simulation test of landing	6	126	
To extend landing legs shortly before touchdown that are able to absorb shock impact, including a locking mechanism to keep the legs in place	One or more legs do not extend when necessary	Lander would balance on remaining legs and would eventually fall down towards the center(s) with the missing leg. Since the force of gravity is low on Enceladus' surface, there is a possibility that there would be minimal damage at worst to the lander	8	Computer system breaks down	2	Construction of rover body to shield computer system from debris	Simulation tests for body shield impact resistance	6	96	
	Legs do not withstand shock impact from landing	In the case that the lander is falling with too fast a velocity, it is important for the legs to absorb shock impact. If the legs fail this test, the mission could very well be over	8	Landing on the surface with an unanticipated higher velocity	4	Construction of rover body to shield organs from debris	Simulation test of landing, as well as shock resistance examinations on objects of similar make up	6	192	
	Locking mechanism breaks down	Again, the legs are a necessary component of the lander for absorbing the shock impact of a landing with a greater velocity than anticipated. If the locking mechanism breaks, the lander can be heavily damaged	9	Landing on the surface with an unanticipated higher velocity, causing legs to effectively crumble	3	Construction of rover body to shield organs from debris	Simulation tests for extending legs and then returning legs to original state	8	216	Shop or construct locking mechanism that withstands a baseline force that would be higher than expected, and make sure baseline force is met with testing
To construct functioning low gain and high gain omni directional antennas to communicate information from the lander to the team back on Earth through low rate or high r	Antennae breaks during landing	Any hopes of communication with the lander is crushed. Purpose of the mission is defeated	10	Unanticipated shock from landing may cause damage to antennae	4	Construction of rover body to lower shock impact on antennae	Simulation test of landing	6	240	Shop for flexible antennae that absorb shock by shaking
	Connectivity issues between antennae on lander and antennae back on Earth	Tech control group may not receive all pertinent information as soon as the lander retrieves it, and possibly some information may be lost due to connectivity issues	6	Bad signal (either because of distance, interference, etc.) between tech control group and computer system on lander	3	Use of low gain and high gain antennas to directly communicate from lander to the Deep Space Network (DSN)	Communications test between antennae and DSN	7	126	
	Information sent by antennae is not clear	Mission will not be able to report or computer system malfunctions	9	Antennae is damaged during mission or computer system malfunctions	3	Construction of rover body to lower shock impact on antennae	Simulation test of landing	7	189	Shop for flexible antennae that absorb shock by shaking
To effectively and safely store fuel for thrusters throughout duration of the flight	Fuel leaks out of the fuel tank	Hydrazine is flammable, especially in low temperatures. Fuel leakage could severely damage our instruments/lander	9	Fuel tank is cracked by debris	1	Construction of rover body to shield fuel tank from debris	Simulation tests for body shield impact resistance	7	63	Develop benchmark test that explores impact resistance of fuel tank shell
	Fuel runs out before landing	The smooth landing is caused by the thrusters, so running out of fuel defeats the purpose, and possibly the success, of our landing	7	Miscalculation on behalf of the team of either how much fuel is necessary or at what rate the fuel is burned	2	Buy thrusters from established company so calculations are based on previous successful missions	Contact company to discuss fuel and thruster performance	6	84	
	Overheating or adverse conditions on fuel affects usability	Making any fuel unusable, no matter the reason, is a costly mistake that will make landing a more difficult endeavor	6	Fuel tank and thruster body are not sufficient in protecting fuel from outside conditions	3	Construction of rover body to shield fuel from extreme temperatures to be experienced by lander during mission	Simulation tests for body shield in extreme temperatures	7	36	
To successfully use thruster combustion chambers and nozzles to serve as a heat shield to protect the lander from adverse temperatures	Thruster combustion chambers are faulty/detach during journey	The lander is subject to adverse temperatures	7	Material of thruster combustion chambers too weak/ connection to body is weak	3	Guideline on material for thruster combustion chambers	Simulation test of Enceladus	3	63	Test multiple mixes
	Reaction air	The lander is		Material of		Reaction time	Simulation test			

	Nozzles are faulty/fall off during journey	The lander is subject to adverse temperatures	7	Material of nozzles too weak /connection to body is weak	3	Guidelines on material for nozzles	Simulation test of landing	5	105	
To use the computer to check the health of the rover including temperature and power levels for possible readjustment	The computer runs out of power	The computer can no longer operate and perform duties like data storage /communication and modifications	8	Batteries/ power source is depleted	4	Account for extra batteries /power source	Simulation test of battery usage over time	3	96	Use simulation tests to find the appropriate # of surplus batteries/ energy needed
	The computer breaks during landing	The computer can no longer store/process data and command adjustments in temperature /power levels	8	Debris and impact from landing causes damage through body of lander, breaking computer	3	Body material specification	Simulation test of landing	5	120	Test multiple times
	The computer disconnects with ability to check temperature and power levels	The computer can no longer decide how to proceed with adjusting temperature /power levels	6	Connection between computer and other systems cut off physically	3	Reinforcements of connectors	Durability test of connectors	3	54	Ensure overall testing of the lander and individual testing of reinforcements
	The computer disconnects with other parts needed to adjust temperature and power levels	The computer is no longer able to store/process new data found	8	Connection between computer and other systems cut off physically	3	Reinforcements of connectors	Durability test of connectors	3	72	
To construct a strong rover body that will protect organs from damage and keep the rover at functioning temperature throughout the journey	The rover body breaks during landing due to approximated impact from flying debris or ice	The rover potentially is unable to collect itself and loses functionality	8	The impact from landing/ debris breaks the body of the rover	5	Body material specification	Simulation test of landing	5	200	
	The rover body breaks during movement on Enceladus	The organs are subject to adverse temperatures /debris causing potential damage in usability	6	Impact from slippage, debris or collisions with particles causes breakage in body	4	Body material specification	Simulation test of landing	5	120	
	The rover body does not keep organs at functioning temperature as hoped	Potential damage of parts/unable to power back on	6	Material of rover body is too weak	3	Body material specification	Testing of rover body	3	54	
To integrate engineering, science and descent imaging cameras to focus on hazardous navigation, taking color images and footage of the terrain, analyse elemental composition of minerals through laser, camera and spectrometer as well as detecting movement of spacecraft for safe descent	The cameras break during landing	The camera can no longer take high quality data from Enceladus	7	The impact of landing/debris breaks cameras	4	Camera cover material specification	Simulation test of landing	5	140	Ensure camera covers have reinforcements and have good functionality
	The cameras disconnect and do not send information back to the rover's antennas	The lander can not send data back to the team on Earth to relay information	8	The connection between cameras and antennas is cut off or disrupted	3	Reinforcements of connectors	Simulation test of connectivity	3	72	
	The camera lens gets clouded/ broken	The lander can not record data accurately for analysis	7	Particles/debris breaks camera lens	4	Reinforcements to camera lens	Camera lens durability test	4	112	
	The spectrometer of the science camera stops functioning	The lander can not take samples of element compositions of minerals	6	Particles/debris breaks spectrometer	4	Reinforcements to spectrometer	Spectrometer durability test	4	96	Take data on what parts are malfunctioning in the test for potential future reference
	The descent imaging camera does not detect movement well	The lander is not able to modify landing if need be	7	The descent imaging camera has some bugs in software	3	Software guideline specification	Descent imaging camera testing	3	63	
	The laser of the science camera inaccurately targets minerals	The spectrometer is unable to get correct, accurate testing of mineral samples	6	The implementation of the laser during integration is faulty	3	Science camera laser reinforcements	Testing of science camera	3	54	
To ensure seamless integration of all parts throughout design and manufacturing of the lander and rover	The parts do not fit together during integration	Connection between parts are potentially lost, the lander does not hold up together	7	Some parts were made too big/ too small	3	CAD Modelling	Small prototype scaled testing	5	105	Do not make the prototype too small or too big
	Some parts easily break under weight of other parts	Some parts of the lander malfunction or stop functioning	7	Some parts were made not taking into account how it will be integrated all together	3	Prior CAD Modelling	Small prototype scaled testing	5	105	

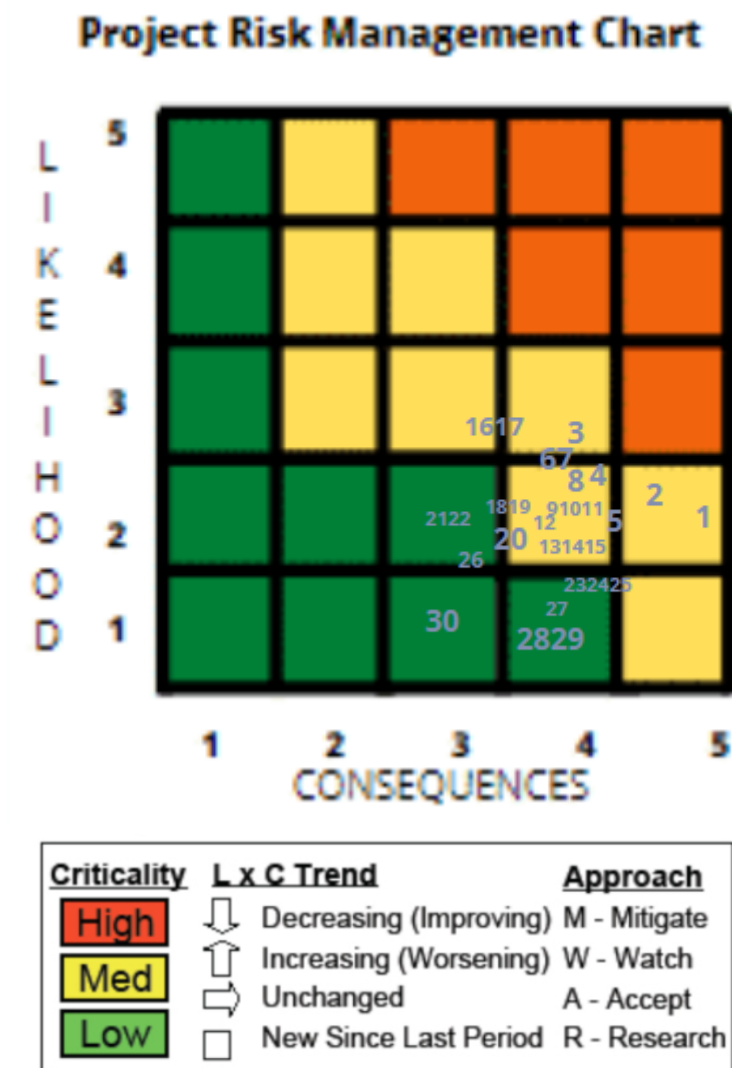


Figure 5: Risk Mitigation Chart

Table 1: Risk and Trends Table

Rank & Trend	Risk ID	Approach	Risk Title
1	R-8	M	Antennae breaks during landing
2	R-7	M	The locking mechanism breaks down
3	R-16	A	The computer runs out of power
4	R-19	M	The computer disconnects with other parts needed to adjust temperature and power levels
5	R-24	A	The cameras disconnect and do not send information back to the rover's antennas
6	R-23	R	The cameras break during landing
7	R-25	A	The camera lens gets clouded/broken
8	R-17	R	The computer breaks during landing

9	R-27	R	The descent imaging camera does not detect movement well
10	R-29	M	The parts do not fit together during integration
11	R-30	M	Some parts easily break under the weight of other parts
12	R-28	M	The laser of the science camera inaccurately targets minerals
13	R-20	R	The rover body breaks during landing due to approximated impact from flying debris or ice
14	R-10	M	Information sent by antennae is not clear
15	R-2	M	Computer system malfunctions
16	R-26	M	The spectrometer of the science camera stops functioning
17	R-21	R	The rover body breaks during movement on Enceladus
18	R-4	A	Sensors do not report information regarding temperature, texture, vibrations, atmospheric particles; or information is inaccurate
19	R-15	R	Nozzles are faulty/fall off during the journey
20	R-14	R	Thruster combustion chambers are faulty/detach during the journey
21	R-9	A	Connectivity issues between antennae on lander and antennae back on Earth
22	R-18	M	The computer disconnects with the ability to check temperature and power levels
23	R-1	M	One or more thrusters fails to work during deorbit or landing
24	R-3	R	Connectivity issues prevent communication lines between the computer system and technology control group
25	R-5	M	One or more legs do not extend when necessary
26	R-22	R	The rover body does not keep organs at functioning temperature as hoped
27	R-12	R	Fuel runs out before landing
28	R-11	M	Fuel leaks out of the fuel tank
29	R-6	M	Legs do not withstand shock impact from landing
30	R-13	M	Overheating or adverse conditions on fuel affects usability

3.1.7 Performance Characteristics and Predictions

Expected conditions landing and operating on Enceladus include passing through a trace atmosphere of water vapor, operating at a mean temperature of 75 Kelvin, and landing on an uneven icy crust.

The atmosphere of Enceladus has little data, but it is known to be thin, so an underbelly that is made to withstand higher temperatures and harsher conditions than otherwise would be necessary to deal with a worst-case scenario. NASA 398 aluminum can maintain high mechanical strength at temperatures up to 370 degrees C, which should be higher than atmospheric heating heats the lander [17]. There should be no need for additional heat shielding other than some insulation of parts within the body that would not respond well to elevated temperatures. Some of this insulation will be needed regardless of heating during landing, as many parts will need insulation from the cold once on the surface. The thrusters will not be able to have the same kind of protection because they need to be able to fire during the entire descent, but they should not need any protection. The thrusters will be continuously firing during entry which will protect them from atmospheric heating plus they are already built to withstand temperatures of up to 800C of burning hydrazine [18]. The effects from atmosphere heating during entry will probably be marginal, however, so this will most likely be one of the smaller issues the lander will face. The thin atmosphere also brings little to no weather. There will be jets of water emerging from the tiger stripes, but we will land far enough away that these will not impact the craft heavily.

The low temperatures on Enceladus' surface will not impact the propulsion system heavily. This is because the thrusters will have been firing in space and will continue to be lit during descent, so the thrusters themselves will be warm from burning the propellant, plus they have their own heaters for where it is needed. The thrusters should, as a result, be able to continuously fire nominally, and provide a slow and powered descent onto the surface. Low temperatures may affect other systems of the lander, such as the motors to deploy the landing legs, but these systems will all have the necessary heaters to warm them up to operational temperatures before they need to be used. The lander will be designed to keep the bulk components within the solid body at operational temperatures, so unless all heaters fail there should be little issue with temperature. It will be designed to lose as little heat as possible as well as having redundant heaters in order to eliminate the chances of heating failure.

Landing on an uneven icy surface could provide the challenges of the landing legs getting secure purchase on the ground as well as the thrusters melting or otherwise kicking up material from the surface. Landing legs will be designed with a crush core to act as a type of suspension to, first of all, reduce the landing loads on the lander at touchdown, but also to help auto level the lander. This will help the lander have a more stable footing on the ground. The legs may also be designed with small ridges to help dig in and grip the icy surface to prevent slipping. The thrusters have the potential to kick up material from the ground or melt the surface directly under them. This could damage components of the lander, so the thrusters will shut off about a meter above the surface and allow the lander to freefall the rest of the distance. Here, the crush cores of the landing legs will help reduce loads on the lander, and it should be able to land safely.

3.1.8 Confidence and Maturity of Design

The system overview consists of various components that will serve to ensure that the descent system and lander design operate successfully. In order to ensure the security of the design, the descent and lander systems will be equipped with state-of-the-art technology that is designed and approved by various teams of professionals to ensure that all possibilities are accounted for. Once the technology has been manufactured, it will undergo strict testing procedures.

The various components of the lander will all need to be individually tested, as well as the assembled machine. For one, the outer body of the lander will undergo various tests to ensure that it can handle potential obstacles and unforeseen problems, such as an accelerated descent period, various shapes

and forms of debris that may damage the body, extreme temperatures, among a multitude of other possibilities. Most importantly, various descent maneuvers will be tested to counteract the possibility of the system being completely destroyed if the thrusters malfunction in any form. While the thrusters are the primary form of descent, backup measures will be taken to ensure that in the event of thruster malfunction, the lander is equipped with a sufficient form of descent that will continue to safeguard its contents and continue the mission on Enceladus. Through consistent testing, the lander is expected to undergo a successful descent process by way of a capable and rigid lander that is well-equipped to deal with all expected as well as unforeseen issues during the mission.

A major part of the initial design had to undergo major modification once more information about Enceladus' atmosphere was further researched. This resulted in changing the primary lander system to thrusters instead of parachute descent. Additionally, it is expected that the monopropellant hydrazine thrusters be further researched and improved upon before implementation, due to the size and power differences in comparison to the *Curiosity* sky crane. As developments about Enceladus continue to surface, it is likely that the components of the initial design further evolve as more information is researched. As the design continues to evolve, the mission expects additional modifications to be made, especially with the overlap of the scientific instruments in the near future that will require further testing before they are allowed to fully incorporate into the lander design.

Once the final mission concept, system-level requirements, and project management plans (Phase A) are finalized, NASA will conduct a review of the proposal. The process can take up to several months and includes Peer Panel reviews, a recommendation from a Program Officer, and a funding decision before a final outcome is reached [19]. While the team is confident in the validation of the proposal, it is likely that NASA makes suggestions for modification, in which case, the team will conduct additional revisions before moving forward with Phase B.

3.2 Recovery/Redundancy System

The landing process, which includes slowing down the lander vehicle and safely completing touchdown, has been identified as one of the highest-risk portions of the entire mission. Many of the systems and technology that are being carried aboard can potentially be damaged during this final phase before reaching Enceladus' surface, and the effects of such damage could very well undermine the success of the mission. Therefore, our recovery and redundancy systems are focused on technology aiding in the landing process and on collecting scientific information.

For starters, thrusters will help decrease the descent velocity to reduce the impact of touchdown. Due to the reliability of a monopropellant thruster, the team does not expect that extra thrusters will be necessary.

Instead, changes in throttle will balance the lander on the remaining functioning thrusters in the scenario that a thruster(s) breaks down. The throttle may be adjusted between 1.8 to 4.9 newtons, which we expect to be more than enough to balance the lander under the extremely-low gravitational field of Enceladus [19].

Our computer system is crucial in allowing the tech control group on Earth to take charge of certain functions when nearing Enceladus' surface. The importance of the computer system cannot be overstated, therefore, the team has decided that twin computers will be installed inside the rover body in case one of the computer systems malfunction. The Curiosity Rover has twin computers, which allowed the team to work with Computer B while Computer A was being restored. This mission shall follow the precedent to waste as little time as possible and to ensure a successful mission [20]. Accompanying the computer systems will be flexible, strong wired reinforcements connecting the computer system to other devices.

The antennas are responsible for communication lines between Enceladus and Earth. Our mission loses its purpose if we cannot communicate scientific data and findings back to Earth for analysis. Antennas are relatively lightweight, so fortunately there is space within the physical constraints of the mission to include multiple antennas of different styles. At the least, the lander body would include one ultrahigh-frequency (UHF) antenna and two long-range antennas. The long-range antennas would include an X band high-gain (HGA) and an X band low-gain (LGA). The purpose of having different forms of antennas is to open as many lines of communication as to effectively control the lander systems from Earth [21].

When landing, the team must ensure the touchdown spot is safe for the lander. Information about the landing zone within the proximity of the descending lander is delivered by the engineering, science, and descent imaging cameras. These cameras will be found outside the shell of the lander body, so the risk of damage is higher for them due to exposure. The chance of damage is relatively small, and the cameras would theoretically not go out until the lander is relatively close to Enceladus' surface (at a point when the team would have a good approximation of the touchdown area). However, the other sensors within the protective shell of the lander body would serve as backup receivers of information. These sensors, while picking up different information compared to the descent cameras, would provide us with valuable insight about the touchdown area to land in the most effective spot.

The fuel tank is crucial to storing the lander's energy source. Therefore, extra fuel will be located in the main fuel tank along with batteries in case the lander uses up calculated fuel for the trip or encounters any issues. The batteries will help charge the computer and will be used by the cameras as well.

Enceladus' atmosphere is rather extreme in temperature and in debris shooting up from the geysers in the Tiger Stripe region. An inner shell, along with a protective outer shell, will help pad its organs from the extremities mentioned. In the case the outer shell cracks, the inner shell would do an adequate job keeping the organs from being exposed to adverse conditions of Enceladus. Padding

around the organs would also provide extra relief if the lander undergoes an unanticipated crash landing.

Nozzles serve as heat shields to protect the lander from adverse temperatures from the thruster combustion chambers. The nozzles of the lander are secured through the nature of design and physical securing. Material specification, simulated testing, and review with external parties are factored in to ensure a successful mission and strong durability.

The spectrometer aids the science cameras mounted on the rover's body in examining close-ups of mineral structures and textures. To prevent scenarios of the spectrometer breaking or malfunctioning, it will be secured with reinforcements because data collection is a crucial part of the mission. Sensors along the outer shell of the rover and high-quality footage through the cameras would also provide valuable information in the case the spectrometer is damaged.

3.3 Payload Integration

The first instrument that is mounted is the camera. This plays an essential role in the landing maneuver, as it provides visual feedback on the landing sites as well as information for the computer on how exactly to touch down. The camera module also houses other essential sensors like the lidar system which tells the computer exactly how far away the ground is, as well as accelerometers and gyroscopes that help orient the lander. This sensor box is mounted on the floor of the lander's body, looking down. This will allow pictures to be taken of the descent as well as the surface of Enceladus, as well as having the sensors work effectively towards the safe landing of the craft.

The next instrument is the infrared imager and spectrometer. This will be modeled after the Ralph instrument used in New Horizons and will look outwards through a cutout in the lander's frame. The telescope will fit in the hull looking out at the landscape so it can analyze the composition and other details of the land.

Finally, the gas chromatography-mass spectrometer will fit inside the lander body. This needs samples from the surface in order to collect data, so a drill will take cores of the ice and bring them into the lander for analysis. The drill is mounted inside the lander and using a motor to lower it, a telescoping array of pipes lowers towards the surface. Once the drill hits the surface, it drills a core of ice, then retracts back up into the body. Once inside the body, the sample is transferred from the drill into the GC-MS as seen in the figure.

Below, figure 6 shows all these parts and how they fit together in an exploded view of the lander.

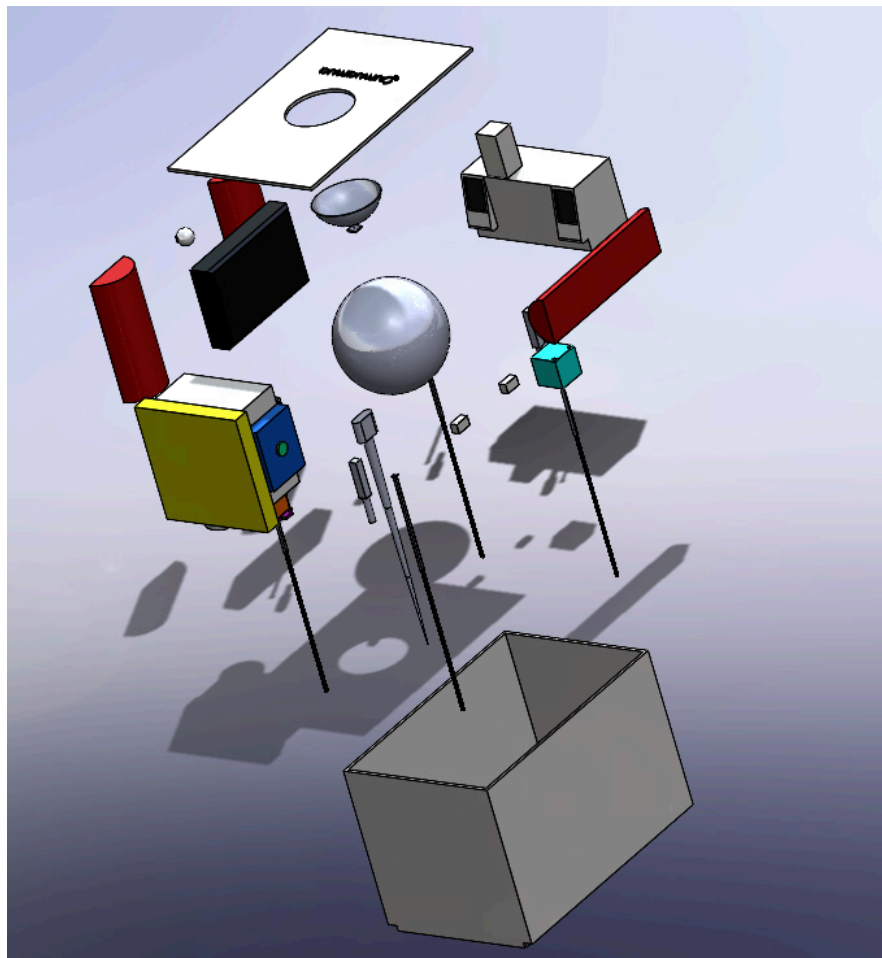


Figure 6: Exploded view of lander.

4. Payload Design and Science Instrumentation

4.1 Selection, Design and Verification

4.1.1 System Overview

For the power system there will be lithium batteries that will power the entire spacecraft and will provide power to all the systems on board. The batteries will be able to carry 40-amp-hour at 12V which will result in 480 watt-hours available to use. For our communication system we will be using an antenna to communicate with NASA's Deep Space Network (DSN). The thermal system is composed of materials and coatings chosen to minimize conductive heat loss as well as radiative heat loss from the lander. The total heat loss after these measures is counteracted by electrical resistance heaters that are controlled by the lander's computer and temperature sensors in the hull. Most of the systems connected to the electronics will be used very intermittently, partly to save battery power, and partly because there will not always be new data to collect. These instruments will be turned on to do measurements when wanted, and then they will be shut down. The GC-MS draws around 65W of power but can draw a peak of up to 154W. The drill used to collect samples for the GC-MS will be used as intermittently as the GC-MS.

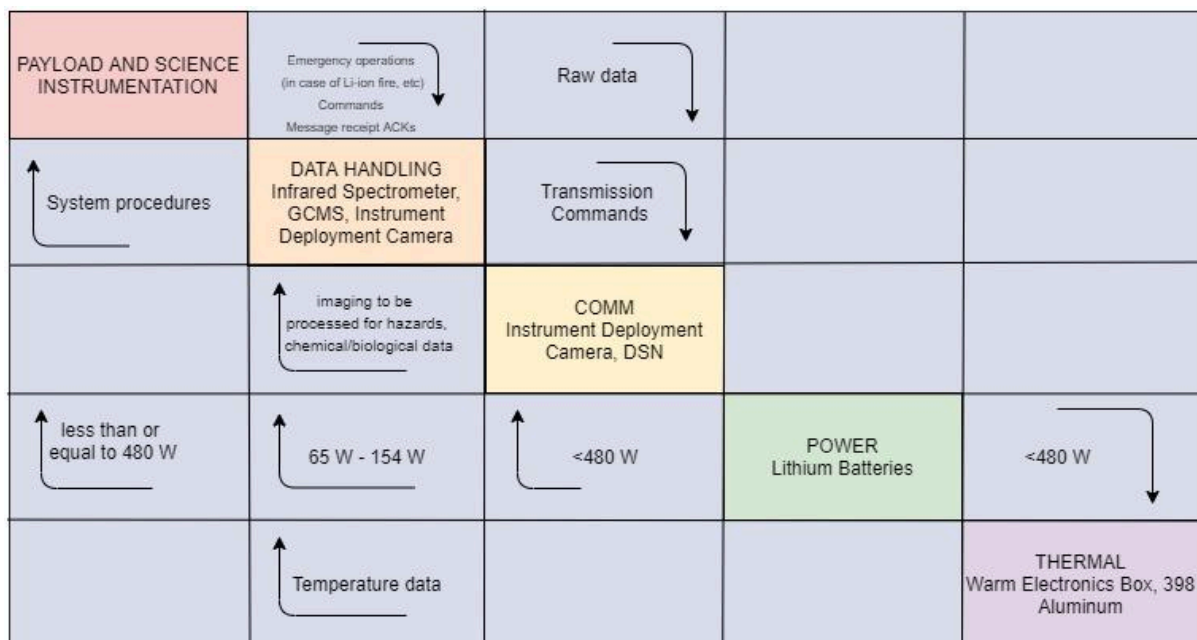


Figure 7: N2 Chart

4.1.2 Subsystem Overview

One of the main components of the science instruments is the Gas Chromatography-Mass Spectrometry instrument. This instrument, also known as GC/MS, begins with the gas chromatograph, where the sample is volatilized. This vaporizes the sample and turns it into the gas phase, then it separates its components using a capillary column packed with a stationary solid phase. As components become separated, they elute from the column at different times, which is generally referred to as their retention times. After components leave the GC column, they are ionized by the mass spectrometer using electron or chemical ionization sources. Ionized molecules are then accelerated through the instrument's mass analyzer, which quite often is a quadrupole or ion trap. It is here that ions are separated based on their different mass-to-charge (m/z) ratios. The final steps of the process involve ion detection and analysis, with compound peaks appearing as a function of their m/z ratios. Peak heights, meanwhile, are proportional to the quantity of the corresponding compound. A complex sample will produce several different peaks, and the final readout will be a mass spectrum.

Using computer libraries for mass spectra for different compounds, researchers can identify the quantitate of unknown compounds and analytes. GC is the separation technique of choice for smaller and volatile molecules such as benzenes, alcohols, and aromatics, and simple molecules such as steroids, fatty acids, and hormones. This instrument is involved in the data gathering component of the mission. Specifically, analyzing incoming information and processing data for further investigation.

In order to collect samples of Enceladus' ice for the mass spectrometer, a drill will be lowered to the surface of the moon. This drill will require a motor to spin the drill into the surface and a motor to raise and lower the telescoping mechanism to and from the surface. This sample will be retracted into the body and moved into the loading bay of the mass spectrometer.

Attached to the GC/MS will be a heating source in order to keep the temperature of the device at optimal operational temperature, which is 233K-293K.

Another component needed in the science instruments is an Infrared Spectrometer. During this process, the interaction of infrared radiation with matter is measured using absorption, emission, or reflection. The method of infrared spectroscopy is conducted with an instrument called an infrared spectrometer which produces an infrared spectrum. An IR spectrum can be visualized in a graph of infrared light absorbance or transmittance on the vertical axis vs. frequency or wavelength on the horizontal axis.

Infrared spectroscopy exploits the fact that molecules absorb frequencies that are characteristic of their structure. These absorptions occur at resonant frequencies, i.e. the frequency of the absorbed radiation matches the vibrational frequency. The energies are affected by the shape of the molecular potential energy surfaces, the masses of the atoms, and the associated vibronic coupling. The resonant frequencies are also related to the strength of the bond and the mass of the atoms at either end of it. Therefore, the frequency of the vibrations are associated with a particular normal mode of motion and a particular bond type. The Infrared spectrometer is also involved in data gathering and analyzing incoming information during the mission.

For power generation, a lithium battery will be used. A battery is made up of an anode, cathode, separator, electrolyte, and two current collectors (positive and negative). The anode and cathode store the lithium. The electrolyte carries positively charged lithium ions from the anode to the cathode and vice versa through the separator, the movement of the lithium ions creates free electrons in the anode which creates a charge at the positive current collector. The electrical current then flows from the current collector through a device being powered to the negative current collector. The separator blocks the flow of electrons inside the battery. While the battery is discharging and providing an electric current, the anode releases lithium ions to the cathode, generating a flow of electrons from one side to the other. When plugging in the device, the opposite happens: Lithium ions are released by the cathode and received by the anode.

Some advantages of lithium batteries are that they are more reliable than older technologies such as nickel-cadmium and they do not suffer from a problem known as the "memory effect" (where nicad batteries appear to become harder to charge unless they're discharged fully first). Since lithium-ion batteries don't contain cadmium (a toxic, heavy metal), they are also better for the environment. Compared to heavy-duty rechargeable batteries (such as lead-acid ones used to start cars), lithium-ion batteries are relatively light for the amount of energy they store. However, some disadvantages of lithium-ion batteries involve safety. Li-ion batteries will catch fire if they are overcharged or if an internal malfunction causes a short circuit; in both cases, the batteries heat up in what is called a "thermal runaway," eventually catching fire or exploding. That problem is solved with a built-in circuit breaker, known as a current interrupt device or CID, which kills the charging current when the voltage reaches a maximum, if the batteries get too hot, or their internal pressure rises too high. But

there remain concerns and, in 2016, the International Civil Aviation Organization officially prohibited shipments of lithium-ion batteries on passenger planes because of the potential danger.

Another instrument for data gathering and communication will be the camera system. This will involve multiple cameras including wide angle cameras underneath the lander that are used to help navigate during descent, as well as cameras facing outwards from the sides of the lander to image the terrain of Enceladus. Information received from the camera will help engineers and scientists guide the deployments of the arm probe for further data collection.

4.1.3 Manufacturing Plan

Another critical aspect to be addressed for the development of the mission is how the scientific instrumentation will be manufactured. This stage will take place between Phase C and Phase D of the mission, thus starting the fabrication process on November 1st, 2028. There is a wide range of possibilities that can be looked upon, ranging from Commercial Off The Shelf (COTS) instrumentation, home made designs/manufacturing or outsourcing to other shops. All of these possibilities have been taken into account when deciding which one would work best for the selected instrumentations and the purpose of the mission. As mentioned in Section 1.4, the main instrumentations to be carried in the payload are a Gas Chromatography-Mass Spectrometer, an Infrared Spectrometer, Lithium Batteries and an Instrumentation Deployment Camera.

The manufacturing of the Gas Chromatography-Mass Spectrometer will be manufactured separately, similarly to how the Mars Organic Molecule Analyser was done. The main reason for this is that normal GC-MS, such as the ones used on the Viking Landers and Curiosity rover, are mostly destructive and only allow the detection of volatile molecules to be detected through the GC. Therefore an improved GC will be designed for this mission to account for the latter issues. When developing the GC for MOMA, two french institutes were chosen to develop the GC instrument since the mission was led by the European Space Agency. For the purpose of our mission, we will partner with Massachusetts Institute of Technology to develop a new GC that will address all issues from older versions. This institute was chosen due to their innovative development of their MEDUSA Gas Chromatography system that was successful in being the first automated GC-MS instrument to be deployed long-term at remote stations whilst also having a high reliability. Therefore, they will be given financial support to continue their work and manufacture a GC for this mission. On the other hand, the Mass Spectrometer and main electronics will be manufactured in NASA's Goddard Space Flight Center. Here a MS instrument similar to the ones used in previous missions will be developed.

When assessing how the Infrared Spectrometer will be manufactured, it was decided that we would develop it within the NASA organization. The team will take Ralph (developed by NASA) as a reference and develop a new one with updated technologies that were not available back when it was designed. This latter instrument is a visible infrared imager and spectrometer developed for the unmanned New Horizons spacecraft, which was launched in 2006. The specifications for Ralph lie close to the ones needed for our mission, therefore NASA will take this instrument as a base reference and develop a newer version with better camera optics.

How the lander will be powered is another key aspect in the development of the mission and has already been deeply discussed throughout this project. It was decided that the best option for the lander was to be powered by lithium batteries that offset the decaying charge by utilizing solar arrays while travelling through space. This instrument would supply enough power to the lander and all instrumentation for the duration of the mission, whilst also adjusting to the mass and volume constraints applied to the lander. The batteries will be manufactured by Yardney Technical Products Inc., based in Rhode Island. This company has supplied Lithium Batteries to other NASA missions such as the Phoenix Mars Lander and the Spirit and Opportunity Mars Exploration Rovers, which

proved to be successful. Therefore, due to the high reliability and good relationship with Yardney Technical Products, they will provide us with the power source for our lander.

Finally, Lockheed Martin Space will be the company awarded with the contract to manufacture an Instrument Deployment Camera. This company was chosen because they developed a similar camera for the InSight lander designed to study the deep interior of Mars, which also proved to be successful. Therefore, Lockheed Martin Space will develop this camera system on their Denver location under the supervision of NASA. Once the camera system is manufactured, it will be sent to NASA in order to accommodate it into the lander.

4.1.4 Verification and Validation Plan

During the Verification and Validation process the team will stimulate the lander's new environment and various environmental changes that will occur during the mission. This will ensure that the lander and all scientific instrumentation was built correctly and is ready for the expedition. For this reason, the plan is to perform extensive testing in many conditions that can range from temperature testing to vibration and other part performance testing. The goal is to stimulate any challenging circumstances that the spacecraft could encounter during its mission and see how the instrumentation stands against these adversities.

First, the Gas Chromatography-Mass Spectrometry instrument will be tested. This particular instrument has a variety of jobs. The team will test it to ensure that it can readily and accurately segregate complex compounds found in water or air. Less volatile chemicals move slower than more volatile chemicals and are therefore separated by a GC/MS. It will also be tested to make sure it can measure the amount of every chemical present in a given sample simply by comparing the particular chemical to pre-measured standard. Lastly, it will be tested to verify it can identify any unknown chemical in a sample through a variety of experiments. The mass spectrometry analysis aids in determining the chemical's unique structure or its "fingerprint" which could be compared to an extensive library of identified chemical fingerprints.

Another instrument that will need to be tested is the Infrared Spectrometer. The team will test the instrument to ensure it can measure the change in dipole moment of molecules due to irradiating them with light at frequencies that trigger transitions between vibrational energy levels. This will be done through a series of experiments.

The lithium battery will also be tested beforehand to ensure it has the capability and power to generate energy for the spacecraft throughout the duration of the mission. There is a wide range of tests that will be conducted. For example a voltage test will reflect state-of-charge in an open circuit condition. Ohmic tests will measure internal resistance and identify corrosion and mechanical defects when high. Full cycle tests will determine the full cycle of charge to discharge to read the capacity of the chemical battery. This provides the most accurate readings and calibrates the smart battery to correct tracking errors. Another test is Coulomb counting which provides coulomb count that relates to SoH. A rapid-test includes time domain by activating the battery with pulses to observe ion-flow in Li-ion and frequency domain by scanning a battery with multiple frequencies. Advanced rapid-test technologies require complex software with battery-specific parameters and matrices serving as lookup tables. SOLI, otherwise known as State-of-Life-Indicator, estimates battery life by counting the total coulombs a battery can deliver in its life. A new battery starts at 100%; delivered coulombs decrease the number until the allotment is spent and a battery replacement is imminent. The full scale is set by calculating the coulomb count of 1 cycle based on a manufacturer's specification and then by multiplying the number with the given cycle count.

Lastly, the Instrument Deployment Camera will be tested. It will undergo "shake and bake" testing to ensure it can withstand the vibrations and other movements it will be subject to. The instrument will be placed on a vibration table and will be shaken intensely. It will also undergo extreme temperature testing for environmental testing. To be able to mimic conditions in space, the

lander will undergo solar thermal vacuum testing where it will be exposed to extremely low and extremely high temperatures, but mainly focused on low temperature as Enceladus is extremely cold. The testing for low temperature will be at least -57 K and up to -127 K to ensure the lander is able to keep its body warm and functioning. Another key test that the camera will need to complete is the testing of its telecommunication equipment for electromagnetic compatibility. It is of utmost importance that the instrument can freely transmit and receive data without any interference.

4.1.5 FMEA and Risk Mitigation

As we continue to investigate and analyze our mission protocols and objectives, we are determining more ways of how our lander could mitigate certain outcomes. Within this section, we will discuss several risks such as being hit by external objectives, losing signal to spacecraft, temperature control and much more.

Traversing around Enceladus will be an important aspect for all of our experimentation. It is important to investigate the possible ways that the lander could be trapped, damaged or stopped due to environmental factors. Different types of damage to the infrastructure's body frame could impact the operational factors that make the lander move. The first aspect that we investigated was the motors that will automate the parallel activation of all motors / wheels so that the lander can move. Some environmental factors that could cause an issue within this scope are external fragments such as dust or largely frozen obstacles, cosmic rays that could interfere with the communication panels between the system body and each microcontroller. We can avoid most of this by applying a shield type, specifically one that doesn't obstruct the lander's ability to move but only give it a protective layer from external sources.

Celestial objects are a big factor when it comes to analyzing external environmental conditions. As we know, Enceladus is exposed to the large universe and is orbiting its parent planet Saturn. Saturn's strong gravitational attraction could be an issue in pulling in external bodies that Enceladus intercepts. To add a protective layer, we could investigate either shielding critical components from small types of fragments or a light umbrella-like structure to shield the probe entirely.

Keeping the lander's temperature at an operational state is important as this is a critical objective to achieve since many instruments require a certain range to operate at maximum efficiency. To improvise and overcome these requirements, we can utilize other instruments that exhibit the counter effect or releasing heat by actions to encapsulate the lander and form a precise range of temperature controls. Enceladus is a very cold moon and is frozen from the outer sheath surface, but as we venture down the planet we start to encounter liquid water which is accumulated from the frozen ice layers above. We can utilize a Warm Electronics Box or lander's Body (as NASA described) to encapsulate the lander with material such as Solid Silica Aerogel. This type of insulation keeps the internally generated heat within the body's system and thus keeps the system functioning at its required temperatures.

Since we haven't explored the surface of Enceladus prior to this mission, other than the satellite images supplied by the Cassini Mission. We can assume that since the surface is icy and solid, that the lander would have an impermeable surface to "Glide" through its checkpoints. The only concern with this objective is that the surface could have artifacts that arise from the irregular compression of the moon which is caused by the strong gravitational attraction of Saturn. As we know, the north and south poles of Enceladus contract and squeeze the inner layer / surface which in turn causes heat to be generated below the surface. This could also be the reason why Enceladus has an oceanic world to explore. To overcome these, we have researched ways that the lander could examine its traversable path by utilizing vision assistance / guidance to see if it may continue in its selected path or there may be obstacles approaching ahead. This is one way of validating if the path is clear of debris, but we could also utilize echo sonic sensors or ultrasonic distance sensors to see if

there is an object in the near distance. The downside of using these types of sensors is that if the target structure is non- perpendicular. Another technology that operates similarly to these distance sensors is a laser based integrator that aims to send a beam towards an object and based on the time elapsed for it to return, this would tell us if there is an object in the near premise.

The instruments on board of the lander will be utilized to experiment on specific tasks per objective. The current risk associated with each instrument is that if any fail to complete their objective to the full extent, we might not be able to proceed with them. To counter this outcome, we will look into methods to maximize other instruments so that their project objective can contribute to another if such instruments have failed. This could include instruments that have the ability to dynamically alter their objective so that it suits different types of occasions. Instrumental failure can be controlled to an extent, but having the ability to substitute a similar objective in turn of the one that failed, this could allow us to still accomplish our task goals as well as dramatically reduce the total possibility of complete failure.

Table 2:Risk Analysis - FMEA

ID	Summary	L	C	Trend	Approach	Risk Statement
1	Rover doesn't traverse its enviroment as expected.	1	5	→	A	If the rover has any mechincal issues during its time on Enceladus, it would be difficult to repair it remotely so that it could traverse to its designated locations.
2	Rover gets hit by celestial object	1	4	→	A	If the rover gets hit by an object, it would entirely depend on the area of impact to estimate if critical system functionality has been compromised.
3	Rover tempreature range exceeds limits	3	4	↓	W	If the rovers tempreature exceeds the required range set by our requirements, it would require us to supply heat from local instruments to keep the critical systems operating.
4	Rover becomes set in a an unoperatable position	1	5	→	A	If the rover were to flip over or stay on on its side, this would result in a mission critical failure that would not allow us to continue experimentation UNLESS our instruments can operate given a non linear operational environment such as a robotic arm.
5	Rover loses signal from spacecraft	2	2	→	W	If the rover loses its communication with the orbitiang spacecraft, we need to find a way to store all future findings locally and attempt to re-connect every so often. Once a stable connection has been localized, we can then continue to transmit all of the stored information and proceed with next objectives.
6	Rover instrument becomes non-operatable	1	4	→	A	If the instrument on board becomes non-functional, we can still attempt and utilize other local instruments to accompany a partial task to complete the failed instruments objective. This would allow us to increase the likelihood of a success given such a failure risk but at the same time decrease the likelihood of a failed mission objective.
7	Rover loses a piece of its motors	1	2	→	M	If the rover loses a tire or a critical motor piece, we need to have the weight distribution equal so that it can still operate with less traction but still complete its imperative tasks.

Table 3: Risk Mitigation Chart

L I K E L I H O O D	5					
	4					
	3				3↓	
	2		5→			
	1		7→		2→, 6→	1→, 4→
		1	2	3	4	5
		CONSEQUENCES				

4.1.6 Performance Characteristics

The lander's projected launch date is April 16, 2044, springtime. Enceladus has a year round average temperature of -330°F, or approximately 72 K, so the time of year does not reflect much on the expected weather conditions for the lander, regardless of its planet Saturn having seasonal temperature differences. The expected conditions on Enceladus at the time of landing are comparable to nearly eleven times that of a freezing tundra biome on Earth, thus it is important to ensure that the instruments in our lander are able to survive such conditions.

To work around the hostile temperatures, we have found the optimum temperatures to keep our instruments running. Those temperatures can be achieved through the insulation with the Solid Silica Aerogel used to create the Warm Electronics Box, otherwise known as Rover's Body, mentioned in 4.1.5. The aerogel has a porosity that is more than 90%, along with a thermal conductivity lower than that of air, making the aerogel a highly insulating material. This means that, ultimately, the aerogel effectively inhibits solid and gaseous heat conduction due to that amorphous slender skeleton and unique nanometer-sized pore structure, making it the most useful insulator to aid in the exploration of this ocean world. This strong insulating material will be using the heat generated from the movement and motors of the lander's payload instruments in order to keep everything shielded from the cold without causing an overheating environment for those instruments.

4.2 Science Value

4.2.1 Science Payload Objectives

When developing the framework for the expedition, the science goals were one of the first topics to be discussed since the whole mission would revolve around this. As discussed in section 1.2.1, it was discussed what were the main objectives to address for this mission and all the data of interest to be obtained. As mentioned throughout the project, the main purpose of the mission is to search for different biosignatures and organics that can live in Enceladus. This would enable scientists to determine whether life is possible in this icy moon. Humans have always questioned whether there is some sort of life somewhere else than on Earth. Until now no mission has been able to detect life on other planets, but the data obtained from Cassini's mission has led scientists to believe that it could host some kind of living organism. This makes Enceladus one of the few places within human reach (as of now) where living organisms could be detected. This is why this mission will search for different biosignatures and organics that can survive in Enceladus' environment, enabling us to determine if there is life in Enceladus or not.

Another objective for this mission is to determine whether the heat source from Enceladus originates from a fairly recent heating event or from the beginning of Enceladus' life. This will be done by analyzing the carbon dioxide, ammonia and the temperature of the water vapor and ice particles spewed through Enceladus' plumes. These clues will help scientists estimate the origins of Enceladus' heat source. Knowing how long and consistent the heat source has potentially been would enable us to better understand how old Enceladus. It could also provide a better insight on whether the icy moon could have harboured life at some point in the past, or even in the present.

As mentioned throughout this document, the lander will be facing extreme temperature conditions on Enceladus, which will be a challenge for the design of the lander. Most of the payload instrumentation and apparatus will hold operational temperature ranges from 228K to 328K, meaning that they will not be able to function properly on Enceladus unless they are equipped with some type of heating source. Therefore, keeping all instrumentation under operational temperature conditions is another payload objective. The team decided to keep all instrumentation inside a Warm Electronics Box (WEB), where heaters placed inside the article will ensure that the temperatures range needed is sustained. The WEB walls will reduce the amount of heat loss to the exterior, which will also help reduce the amount of power needed to keep such temperatures.

To ensure the reliability of the data obtained from the payload instrumentation, data will have to be analyzed from five different locations within the Tiger Stripe region in Enceladus Southern Pole. All instrumentation should be able to assess five different samples and send the data back to Earth. This will ensure the reliability of the results provided from all the instrumentation apparatuses.

4.2.2 Creativity/Originality and Significance

As mentioned in multiple occasions, Enceladus is a moon from Saturn where there is limited information available about it. Our concept mission proposes to land in the Southern Pole of Enceladus in order to sample data from the surface of it. This has not been done before in this moon and could lead to the possibility of discovering life organisms outside of Earth. The team is also proposing for the lander to visit 5 different locations where it can sample data, therefore enabling a wider variety of samples in case some issues arise. The Tiger Stripe Region from the Southern Pole was chosen as the landing site and the area of operation for the lander due to their high thermal activity. It consists of 4 parallel stripes that are 130 km long, 2km wide and 0.5km deep. This region can be appreciated in the following figure, where the circle encloses the four tiger stripes.

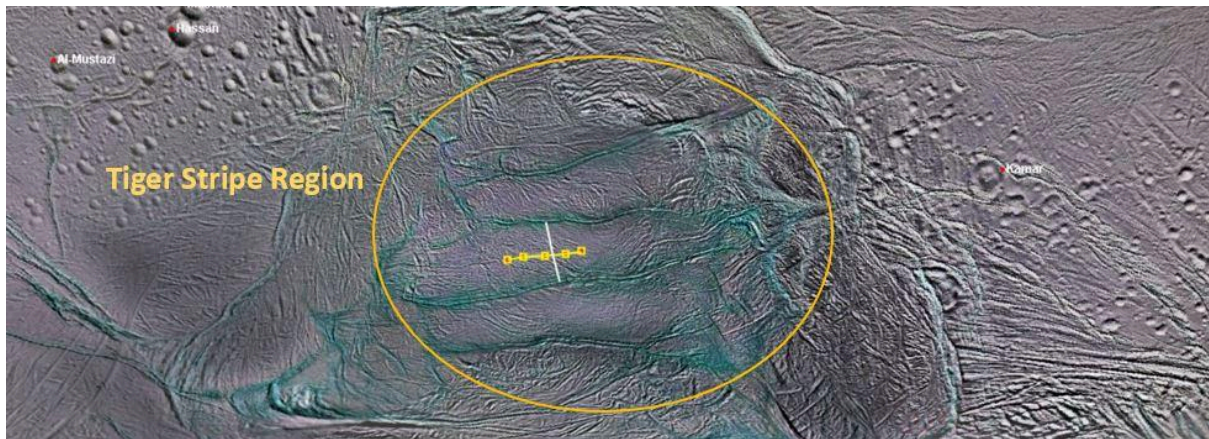


Figure 8: Enceladus Southern Pole Magnified by a factor of 8

This Tiger Stripe Region has been observed to be composed of various plumes spewing water molecules from the liquid ocean beneath it. Therefore, if the lander is able to land close to these stripes, there are high chances that it will be able to recover water molecule samples that have been spewed off through these vents at the surface. As it will be mentioned in Section 5.2.2., the lander should be at a minimum distance of 15 km from any tiger stripe in order to ensure its safety. These plumes exert particles at incredible high speeds that can go hundreds of kilometers into space, so it can be expected to be very adverse conditions close to them. After a careful and in depth assessment, the team determined the landing site and the following data sample sites to be between the Cairo Sulcus and Baghdad Sulcus within the Tiger Stripe Region. The path the lander will follow can be seen in the figure below.

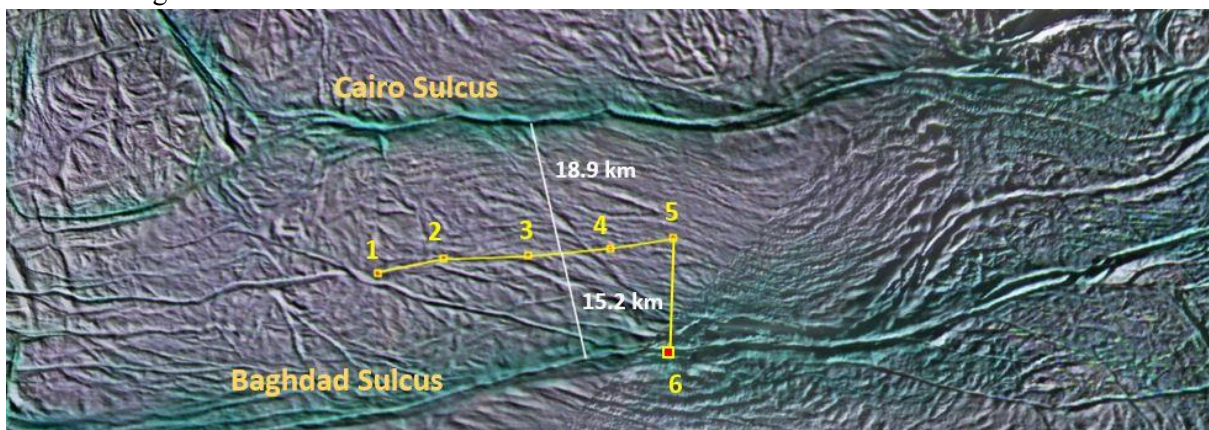


Figure 9: Enceladus Southern Pole Magnified by a factor of 32

This region in between these two stripes is sufficiently large enough to ensure the 15km minimum distance the lander needs to have to ensure its safety. As shown in the image, the closest it will get to one of these plumes is at 15.2km. The lander will travel a total of 37.1 km from the original landing site (labelled as 1) until the final data sampling location (labelled as 5). In the following table the exact coordinates with the distance the lander will have to travel from site to site is supplied.

Table 4: Lander Location Specifications

Location Number	Coordinates	Distance From Previous Site
1	158.03 E, -80.96 N	0 km
2	164.99 E, -82.71 N	8.4 km
3	173.26 E, -84.89 N	10.6 km
4	194.71 E, -86.49 N	10.3 km
5	228.26 E, -87.13 N	7.8 km
6	105.26 E, -89.00 N	15.3km

As shown above, the lander will be able to travel in a linear manner without many abrupt changes in direction, averaging a distance of 9.3 km travelled from one location to the next. The separation between sites was determined after careful calculations done by the engineering sub-team on how far the lander can travel after landing into Enceladus. This will ensure a wide variety of samples from distinct places to be collected.

The team decided to send the lander into the Baghdad Sulcus as the final part of the mission, shown in Location 6 on Figure 9. As mentioned throughout this report, the team decided that there was no real benefit of leaving the lander at location 5 after or the data sampling had been done. Therefore, it was determined that it will be sent into the plumes with the goal of recording all the data possible before it gets destroyed. This information collected could potentially give us a much better insight into the composition of these tiger stripes and the ocean world beneath it.

4.2.3. Payload Success Criteria

The success criteria for the mission include sample data collection from at least five different locations within the Tiger Stripe region of the southern pole of Enceladus. These data samples will then be further investigated and examined. By collecting data from a variety of locations, the results will be more reliable and a biosignature will successfully be identified if it is recorded in at least two locations.

Additionally, because of the challenges associated with landing in a new body within the solar system, another success criteria is the successful transmission of data back to Earth from at least four of the five locations. This criteria ensures that the data will be able to be compared and a biosignature can be identified and organic readings can be compared between our mission and the Cassini mission.

During the mission, communication from two different spots should gather readings on silica particles between the currently known limited size range of two to eight nanometers in radius. Silica particles with radii outside of this range should also be monitored and investigated. This is important in order to rest the idea of heating through rock-water interactions.

A failure that would deter the mission is unsuccessful transmission of data back to Earth. If at any point, the data collected on Enceladus cannot be received on Earth, this will be considered a failure in the mission. The data will need to travel from the spacecraft to the orbital satellite and then to Earth. The core purpose of the mission is data collection, therefore, this step is vital in the ensured success of

the mission. This data is of vital importance due to the unique characteristics of Enceladus. During the Cassini mission, water particles, hydrothermal vents, and organic compounds were discovered on this moon of Saturn. In the search for other life forms in the universe, it is crucial to visit Enceladus. It has the recipe for life and could present unique information about how life forms and how Earth came to be.

Another failure to the mission would be inhibited mobility of the lander. An integral part of the mission is the collection of scientific information unique to the different locations on the Tiger Stripe Region of Enceladus. This information is needed in order to build reliability, consistency, and opportunity. If the lander is unable to navigate the terrain, then less spots will be able to be investigated and less data collection. This would decrease the accuracy of the data resulting in less discoveries being made.

The last failure to the mission would be temperature variants outside of the operational temperature conditions of the lander and the instrumentation onboard. In order to be able to sample and analyze data, the equipment including the Gas Chromatography-Mass Spectrometry instrument, Infrared Spectrometer, and cameras must be kept in certain temperature ranges. If the temperature is not kept within the range, then the instruments will fail to function due to extreme freezing conditions on Enceladus. It will be considered a failure if there is a malfunction in the system and the instrumentation fails to operate optimally.

Due to the presence of plume venting, hydrocarbons in the plume, a salty ocean, and hydrothermal vents on the seafloor, Enceladus has the potential to be a habitable ocean world well beyond Earth's habitable zone. Planetary scientists believe discoveries made on Enceladus have changed the direction of planetary science. Starting in 2005 with Cassini's cameras transmitting the first-ever detailed images of the south polar region of Enceladus, excitement and interest in this ocean world has increased. During a close flyby in 2008, Cassini was able to sample the plume directly and detected a surprising mix of volatile gases, water vapor, carbon dioxide, and carbon monoxide, as well as organic materials. However, the density of organic materials was about 20 times denser than expected.

Enceladus is a long way from Earth. If life were to be discovered there, it would be highly likely that such life had originated there, independently from life on Earth. That would be a very important discovery. It would provide further evidence to suggest that the galaxy is teeming with life, because if life began independently on two different bodies in the solar system, then surely life is also present on many of the potentially habitable planets that are being found around other stars. This mission would help answer a question that many scientists and people have been asking for years and would shape the future of space exploration for years to come.

4.2.4. Experimental Logic, Approach, and Method of Investigation

Collection of data and samples has been established, undoubtedly the preeminent goal of this mission. With this in mind it's important to note how the structure of the lander aids to accumulate and compile fragments and specimens.

All of the samples that will be collected will derive from the surface. The lander will be collecting these samples from the landing point and various other locations on Enceladus. The surface of Enceladus is capped in fresh ice, so the lander is poised with sturdy legs and a drill that unfolds. The drill retracts to feed samples to the main body of the lander.

The drill unravels itself in a manner similar to a telescope. Naturally, this provides stability and accuracy to collect samples of the surface of Enceladus. Once the drill is unleashed it will perforate the surface and collect cores of ice that will then be brought back into the body. Once inside

the body, the sample is transferred from the drill into the GC-MS and analyzed. Conspicuously, the surface of Enceladus isn't mapped out in a way that can make travel across the moon precisely outlined. Therefore, samples will be taken where and when they can.

4.2.5. Testing and Calibration Measurements

Upon reaching the surface of Enceladus and prior to initiating the first steps to validate that every equipment and instrument is operational at its current condition, we must exhibit several factors into each instrument so that its functionality is symmetrical to its intended operations. This phase will occur directly after powering on support systems so that every action that the lander will create must be validated in a simulation environment. This section will elaborate on each scientific instrument on board the lander as well as anticipate a testing ground so that we can verify each instrument is recording samples of data, that being physical or vision-based, to validate the project objectives.

The first instrument to be tested would be the Gas Chromatography-Mass Spectrometry. This type of tool is dramatically difficult to simulate in a local environment as its results depend on the molecular structure of a data sample being given and thus, cannot replicate similar environmental conditions without extensive local data. The best case scenario is to attempt and mimic the simulated substance upon arrival to Enceladus so that we can determine if the same tested results we acquired on Earth are determined to be the same on Enceladus, this would validate that the system is calibrated and functional as intended.

After landing on Enceladus, we will need to calibrate the core instrumentation of our lander and system based configurations such as the navigation system, telemetry system module to validate the lander's current position given its telemetry information to the orbital satellite. Other functionality checks such as system clock reading, this is an important step to validate that the internal circuit structures are processing information and tasks as expected. This was a known effect on electronic devices in space such that specific types of radiation can knock off binary signals within a processor and alter the outcome of a specific instruction. Some ways this has been countered is by having multiple processes run the same instruction at the sametime and one one of the outputs does not agree, then it must have been hit by this issue, then we can swiftly reset that device's binary representation to correct it for future processing.

The next phase of testing and validation after landing on Enceladus is to assess the hardware temperature ranges so that they fall within their acceptable operation zones. This can be achieved by utilizing an insulator to trap in the heat generated from components that radiate heat. We will need to utilize specific heat generators since the temperature on Enceladus is dramatically low and requires extensive power to keep the temperatures in working operations.

The last element that needs to be calibrated would be the camera module, as its lens could be deformed due to the entry and landing onto Enceladus and its low temperatures. We can attempt to have a preset direction to sample several images from and validate that the images being processed are as of such accuracy of the intended objects. Another key factor for imaging is to utilize some type of radar based tracking to have several preset locations of interest for high quality image analysis, as images can relay back the characteristics of the surface of Enceladus as well as close ups to objects.

4.2.6. Precision of Instrumentation, Repeatability of Measurement, and Recovery System

Mass: 10.5 Kg
 Power: 7.1 Watt (maximum)
 Telescope Aperture: 75 mm
 Focal Length: 657.5 mm
 f/#: 8.7
MVIC: Time Delay and Integrate (TDI) and framing arrays
 2 Redundant 5024x32 Pixel Panchromatic TDI CCDs (400 – 975 nm)
 Four 5024x32 Pixel Color TDI CCDs
 Blue (400 – 550 nm)
 Red (540 – 700 nm)
 NIR (780 – 975 nm)
 Methane (860 – 910 nm)
 5024x128 Frame Transfer Pan CCD
 13 μ m x 13 μ m pixels
 Single pixel Field of View: 19.77 μ rad x 19.77 μ rad
 TDI array FOV: 5.7°x0.037°
 Framing camera FOV: 5.7°x0.146°
 Focal plane temperature: <175 K
 Pan TDI rate: 4 – 84 Hz
 Color TDI rate: 4 – 54 Hz
 Frame transfer integration time: 0.25 – 10 sec.
LEISA: 256x256 element HgCdTe array operated in pushbroom mode.
 40 μ m x 40 μ m pixels
 Single pixel Field of View: 60.83 μ rad x 60.83 μ rad
 FOV: 0.9°x0.9°
 Focal plane temperature: <130 K
 Filter segment 1 (1.25 – 2.5 μ m) average resolving power ($\lambda/\Delta\lambda$): 240
 Filter segment 2 (2.1 – 2.25 μ m) average resolving power ($\lambda/\Delta\lambda$): 560
 Frame rate: 0.25 – 8 Hz

Figure 10: Specifications of Infrared imager and Spectrometer

The infrared imager and spectrometer [1] the team has decided to use will be modeled after the Ralph instrument used in New Horizons. In the table above it shows the precision of both parts of the Ralph. Looking at the table above it shows us how accurate each part of the Ralph works, so then the team can expect our instrument to have the same measurements as the Ralph. The team will then verify the data by testing how the instrument works in different conditions, such as what it may encounter on the Enceladus surface.

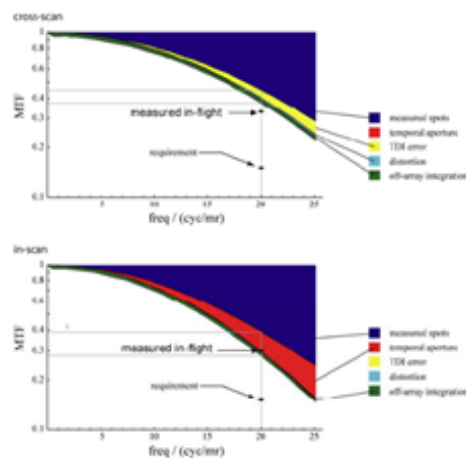


Figure 11: Accuracy of Infrared Imager and Spectrometer

The accuracy of the infrared imager and spectrometer [2] is around $\pm 2^\circ$ Celsius, making the infrared imager and spectrometer very reliable and accurate. ≥ 0.15 at 20 cycles/milliradian. This graph shows its requirement is 15% minimum overall MTF at 20 cycles/milliradian. But it easily exceeds what it was expected. So from this graph the team can expect the infrared imager and spectrometer to work very

accurately. The equation: $MTF(d) = e^{-\left(\frac{d\pi 0.01977 FWHM}{2(\ln 2)\left(\frac{1}{2}\right)}\right)^2}$ can be used to predict the measured spots of what is shown in the graph as well. After the team does their testing on infrared imager and spectrometer, the team can expect it to have the same accuracy and precision of the Ralph.

Precision of Gas Chromatography-Mass Spectrometer

Gas Sampling:	1. Continuous direct atmospheric gas sampling. 2. Batch sampling for GCMS analysis, distributed over descent altitude. 3. Sample enrichment for MS (100-1000× enrichment).
Ambient Pressure Range:	1-1500 hPa.
Altitude Above Surface:	176-0 km.
GC System:	3 parallel columns with H ₂ carrier gas. Independent MS ion source detection.
Mass Analyzer:	Quadrupole Mass Filter.
Ion Source:	Five sources, electron impact ionisation. Maximum operating pressure 1×10^{-3} hPa H ₂ , 5×10^{-4} hPa N ₂ .
Ion Detector:	Dual Secondary Electron Multipliers, pulse counting and analogue current mode.
Background Noise:	1 count/min.
Mass Range:	2-141 dalton.
Sensitivity:	1×10^{14} count/s/hPa source pressure.
Dynamic Range:	$\geq 1 \times 10^8$.
Resolution/Crosstalk:	1×10^{-6} for adjacent half mass up to 60 dalton, less for higher mass.

The table above shows the precision of the Gas Chromatography-Mass Spectrometer.

Accuracy of Gas Chromatography-Mass Spectrometer

Looking at the GCMS that was aboard the Huygens Probe the accuracy of this GCMS the team knows that the transmission efficiency will be 100%. In this article it also states that the instrument sensitivity for 100% ion transmission is approximately 1×10^{14} . In this article it also states the range of detection for ions, so then the team will know approximately when to take measurements as it gets closer to Enceladus

Precision of Cameras

	Navcam	Hazcam
Pixel scale at the center of the FOV	0.82 mrad/pixel	2.1 mrad/pixel
Focal Length	14.67 mm	5.58 mm
f/number	12	15
Entrance Pupil Diameter	1.25 mm	0.37 mm
Field of View (horizontal × vertical)	45 × 45 degrees	124 × 124 degrees
Diagonal FOV	67 degrees	180 degrees
Depth of Field	0.5 meters–infinity	0.10 meters–infinity
Hyperfocal Distance	1.0 meters	0.5 meters
Spectral Range	600–800 nm	600–800 nm

The cameras the team will be using will be modeled after the camera on the Mars Curiosity Rover. On the rover they had two types of engineering cameras: Hazard avoidance camera (HazCam) and Navigation Cameras (Nav Cams). The HazCams each have a 124X124 degrees field of view, and they are able to be used for mapping out terrain. They are able to map out as far as 3 meter in front of it, and 4 meters wide as far at the farthest distance. The NavCams have a 45X45 degree field of view, and work in cooperation with the HazCams to view the terrain. On the Curiosity Rover there is also a Descent Imaging camera which will work alongside the NavCams and Hazcams.

Accuracy of Cameras

The HazCam has an accuracy of ± 5 mm, and for both the NavCam and HazCams [3] we can use the following approximation: for stereo ranging accuracy. When the team does testing for the engineering and descent cameras, the data can be compared with Curiosity's data to make sure its accurate.

Recovery system

In case of failure on the orbiter [4], the team has come up with a plan to make sure the data that was collected is able to be sent back to earth. If an instrument does fail the other systems will pause until that data is sent to earth, and then the other systems will continue. We will have two computers on board so the data will be stored there and then sent back to earth using the antennas using the deep space network. The computer will then try to diagnose why the instrument failed and continue with all other systems. During the whole mission the response team will be checking the systems for failure and looking for the expected data to make sure there are no anomalies.

4.2.7. Expected Data & Analysis

The instrumentation onboard our lander that would utilize data analysis are the Infrared Spectrometer, GCMS, Instrument Development Camera. We will discuss each instrument and its associated data samples so that we could reference what data might be speculated as once it's retrieved from Enceladus.

The Infrared Spectrometer main function is to identify the molecular structure of the given sample by passing an infrared light through and producing a spectrum that defines the molecular type. This is important to be swiftly able to identify geological structure on Enceladus and maybe molecular (If the oceanic world is sampled below the surface of Enceladus). The data retrieved by an Infrared Spectrometer is then processed and analyzed to define the samples element.

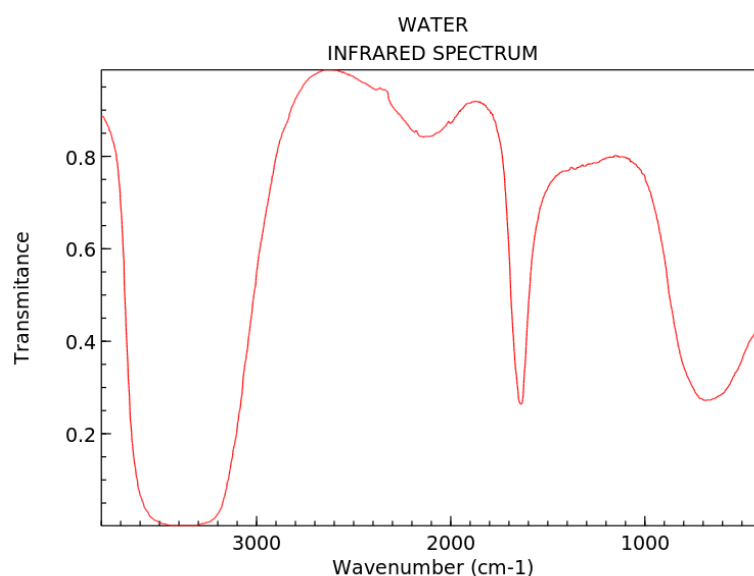


Figure 11: Data Analysis of an Infrared Spectrum for Water (H₂O)

As seen from the figure above [5], we can analyze any sample if processed through an Infrared Spectrometer by applying statistical analysis on the current spectrum and compare it to other known samples. The differential computation between two known spectrums can articulate a factor of confidence that relates the similarities between them and assess the possibility that this acquired result matches the original data, such as Water or specific minerals.

The next instrument to analyze is the Gas chromatography–mass spectrometry or GCMS. This type of instrument is utilized to assess mixtures of compounds and essentially important for us to analyze near the plumes of Enceladus as well as low surface atmosphere gases. Below is an image [6] of how different compounds or mixtures are detected as their bond based representations are displayed. The true data statistic that contributes to the classification of what type of elements are within the mixtures is our core focus. Since we have many data samples of what specific compounds look like under statistical analysis, we can compare and contrast the differences between how several of these compounds relate to predict if they have any relations.

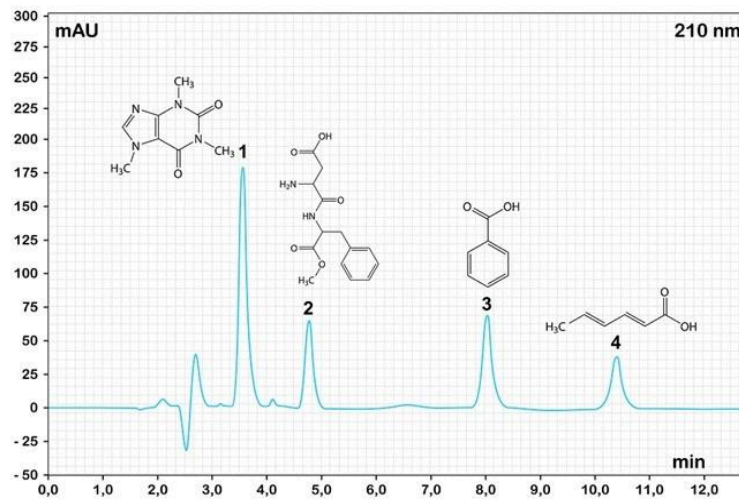


Figure 12: GCMS Analysis of different organic compounds

The last scientific instrument under assessment is the camera module. The data that is perceived from this module doesn't have much initial analysis, but can be utilized to understand the structure / surface of the local environment upon landing. Some types of images that aren't typically classified as imagery data, such as a thermographic camera [7] can detect differential areas of temperature ranges and be able to map areas of which the temperature readings have taken place. This can then be used in Data Science to analyze and attempt to classify why such ranges have occurred, specifically as we approach the Plumes ejection zone to see if the temperature decreases due to the eruption heat radiating away from those areas.

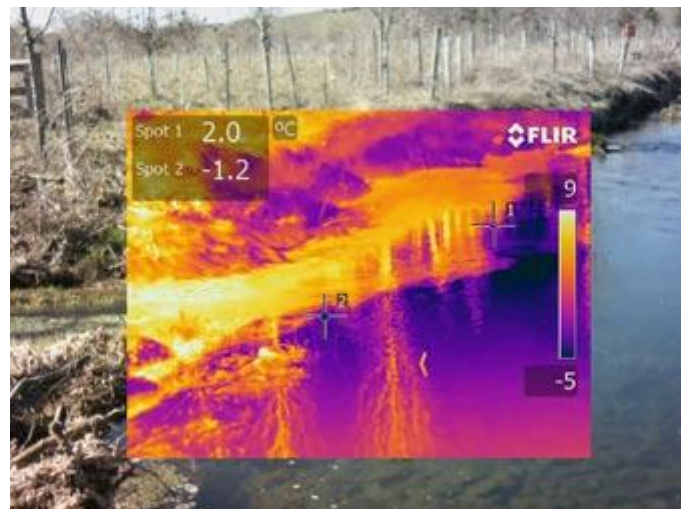


Figure 13: A Thermographic image example of a lake, denoting different ranges of heat radiation.

5-Safety

5.1 Personnel Safety

5.1.1 Safety officer

The safety officer assigned for this mission will be Julian Perez-Doval. Responsibilities for such position include:

- 1) Establishing and assuring compliance with NASA Safety and Mission Assurance (SMA) strategies, policies and standards
 - I. Documenting reports of occurrences
 - II. Recording statistical information
- 2) Reviewing risk identification and assessment methodologies for personnel & workplace hazards
 - I. Conducting inspections & completing audits of worksites
 - II. Reviewing working conditions
- 3) Providing risk mitigation recommendations, and assuring acceptance & awareness of all risk management plans among team members
 - I. Conducting Personnel Training

1

Research undergone to enhance team safety:

- 1) Review of NASA's Office of Safety and Mission Assurance (OSMA) website, including general policies and division-specific tasks for safety
- 2) Quality & safety of facilities and machinery used for mission
- 3) Machinery/lab workshops to certify team members to work with complicated machinery and potentially harmful chemicals
- 4) COVID regulations for the workplace

5.1.2 List of Personnel Hazards:

OSHA's five workplace hazards (safety, chemical, biological, physical, ergonomic) all connect to mission development. Safety hazards include incidents of spills on floors, blocked walkways and emergency exits, loose equipment that could potentially fall on team members and electrical hazards including loose wires [2]. These hazards also include unguarded machinery, moving parts that could cause injury and working in inadequately small spaces. Psychosocial hazards are another concern for teammates throughout the mission, caused by the stress from an overwhelming workload and time constraints.

The biggest chemical hazard is the propellant for the landing thrusters. The propellant is not yet determined, but if it is a monopropellant like hydrazine it would be highly toxic. Another chemical hazard comes from testing the tools that will be a critical feature for data collection of biological signatures on Enceladus. Day-to-day chemical hazards include chemicals in unlabelled containers, fumes from production and working with flammable fuels.

Biologically, the main risk is the coronavirus (SARS-CoV-2) which largely began to affect the United States early 2020. As a result, multiple aspects of the mission must be reevaluated. Currently, the majority of the team's work has been occurring remotely, but it is inevitable in-person meetings will occur, especially in the phase of production. If there is little development in vaccine production

by mid-2021, Phase A will likely be conducted remotely. Because the phase of production is scheduled for in a few years, there is change the coronavirus will have died down by then. If not, the team will be at risk during in-person activities. If mission planning falls behind most scheduled margins and safety nets, the team will revisit the schedule and make appropriate adjustments.

Physical hazards include temperature extremes and exposure to constant noise near machinery [2]. Using machinery such as drill presses, mills, saws, or any other type of tools allows for the possibility of cuts, abrasions, or other physical hazards. Much of the manufacturing may be done with CNC machines or other similar automated processes, but for any work done by hand, there is an inherent risk of personal injury. Most of the lander will be made from aluminum, which means that sparks or metal shavings may fly off of the material and could cause damage to eyes or skin. Aluminum can also get hot while being machined, so there is the possibility of getting burns from touching hot material. For team members who work to assemble outside, physical hazards also include high exposure to ultraviolet rays.

Lastly, ergonomic hazards include frequent lifting and working with heavy equipment leading to poor posture and potential back strain [2].

5.1.3 Hazard Mitigation

OSHA's five workplace hazards (safety, chemical, biological, physical, ergonomic) will be thoroughly evaluated in the workplace. There will be regular investigations and assessments conducted by the safety officer throughout the mission timeline for modification of protocols as time goes on as well as determining if current protocols are being carried out appropriately. Records of previous injuries and incidents will contribute to developing newly improved safety measures and conducting personnel training to team members [3].

Safety measures will be taken to ensure everyday facilities and equipment are safe to use. These measures include keeping corridors and emergency exits clear of hazards at all times, creating procedures for spills, securing any unused equipment down and ensuring day-to-day objects are at an appropriate reach and weight to hold. Lastly, to mitigate electrical hazards, the team will conduct quality assessments on wires prior to usage and throughout mission development. Worn out equipment will be exchanged out. Aspects like fire safety will be understood through mandatory team training prior to the production phase.

In order to mitigate risks associated with handling chemicals, proper safety equipment should be worn. Hydrazine should only be handled with gloves, eye protection, full suit, respirator, and boots, for example. If RP1 is used, it is much less toxic and needs fewer safety precautions [4]. Fuels are flammable and subsequently should be kept away from extreme heat, open flames, exposed electrical connections that could cause a spark, or anything else that could ignite the material. Fuels should be stored in containers that limit chances for combustion. All chemical-storing containers should be labelled clearly to prevent misuse and hazardous materials will be labelled as such.

In the effort to prevent the spread of the Coronavirus disease (COVID-19), all team members shall adhere to all recommended measures as outlined by the Centers for Disease Control (CDC) and the World Health Organization (WHO). This includes: maintaining at least a distance of 2 meters between each member, limiting meetings to at most 10 people in a people in conference rooms with sufficient space for maintaining social distancing, plexiglass dividers where possible wearing a mask at all times when inside the facility, constant sanitation of hands using alcohol hand-sanitizer with a strength of at least 60% alcohol (the facility will have at least 1 hand sanitizing station per room) [5], keeping rooms well ventilated, cleaning and disinfecting surfaces frequently especially those which are regularly touched, performing temperature checks at the facility's entrance, seeking medical attention and self-isolate when feeling any minor-symptoms [6]. All meetings will abide by regional rules and regulations for gatherings during the coronavirus. Team members will be told to stay home

if they feel ill and complete work remotely. Testing of crew will be conducted regularly and team members will abide by quarantine regulations if tested positive. In the event of harmful fluids being exposed in the workplace, first aid will be administered and a prompt evaluation of the area in order for the risk to both people and equipment to be minimal. As of now, the team will continue to conduct work remotely and will ensure that each individual is taking personal responsibility and efforts to safeguard themselves from the effects of the coronavirus.

Physical safety precautions will include regular breaks, standard temperatures, and provided protective equipment for extended periods of loud noise, sensitive fumes, or toxic materials [2]. Radiation exposure is a major component of this section due to the sensitive nature of the potential use of the radioisotope thermoelectric generator. In the event that the team decides to utilize the RTG as the power source, standard safety protocols regarding radiation will be administered. This includes, but is not limited to, contacting an independent organization specializing in radiation to ensure all safety protocols are met [7], using the three principles of time, distance, and shielding [8], as well as having extended oversight during the process. For team members who work outside, breaks will be given regularly along with recommendation guidelines to mitigate against UV rays including sunscreen, protective clothing, etc.

Ergonomically, the team will be encouraged to have good adherence to a work-life balance that ensures that their performance on the team is not compromised, while maintaining a healthy life outside of work that will serve to benefit overall execution of work-related tasks while ensuring that there is not burnout. These practices include encouragement to join hands-on meetings and activities in order for reduced screen-time and to mitigate the harmful effects of extended periods of sitting. The safety officer will determine that team members are not working extended hours or are experiencing physical or mental strain from difficulties regarding mission tasks. In the event this happens, the safety officer will take appropriate measures to ensure that the overall safety and health of staff is not compromised and will take steps to solve any rising issues.

All team members will be required to complete training pertaining to the risks related to the mission [1]. These trainings will be conducted at the beginning of the mission, and specific trainings will be revisited once their content becomes more relevant to the current phase. Training will also be modified during the mission timeline to account for unprecedented safety concerns, such as the most recent example of the coronavirus. If this occurs, the team will be required to revisit the training modules.

5.2 Lander/Payload Safety

5.2.1. Environmental Hazards:

We have identified many environmental hazards from Enceladus that could be critical to our lander/payload systems safety. One of them is Enceladus's ability to eject water vapor or plumes at an intense speed up to 500 m/s [1]. Such particles have the ability to destroy and harm anything that is encountered on its path due to the extreme force at which it is being exerted from the plumes. The surface temperature encountered in Enceladus is another critical environmental hazard to take into account. It is extremely cold (peaking at 72 Kelvin [2]) since it does not account with a thick atmosphere to protect it from space temperatures and composition. These temperatures are on the extreme end and thus our robotic unit must be able to generate internal heat to oppose this effect.

The surface of Enceladus seems to be smoother than our Lunar's surface. Many of the rigids are not as volatile as the ones seen on our Lunar. Since Enceladus is locked facing a single direction on Saturn, some areas are less exposed to the galactic world which leads to a smaller amount of debris impact into its surface. We could furtherly analyze the surface structure by examining the images supplied by the Cassini mission which could reveal the most suitable landing site [3]. The Mission Success focus should also be taken into account when determining a landing position that suits the progress of the mission objective.

Orbital threats such as Asteroids and Rainfall from the Plumes must be considered as Enceladus does not have a strong atmospheric protection as the one encountered on Earth. Since this moon is constantly facing Saturn, there is a greater probability that its outer facing surface has received impacts from space debris, leading to the formation of many craters. On the other hand, the Southern Pole has minimal impacts but a high thermal activity. This does introduce the hazards of Plumes ejecting our probe off course and must be cautioned prior to proceeding. Certain measures could be implemented such as prior examinations / time correlation of how frequent these Plumes occur or if they are on a constant set interval. This could aid the scientific instruments to achieve their gathering requirements without being exposed to these conditions.

At the South Pole, where the plumes are, smaller particles are sprayed into space, however, larger particles fall back to the moon's surface where they accumulate as icy mantling deposits. Galactic Cosmic Rays (GCRs)[5] that strike both the moon's surface and the icy particles produce an ionizing radiation in the form of high-energy electrons, neutrons, protons, gamma rays, and muons. This radiation is in addition to the population of energetically charged particles in Saturn's magnetosphere. We must recall that radiation involves the destruction of chemical bonds and the creation of free radicals. Thus affecting organic matter and destroying biomarkers over time.

Dust adhesion to the payload is also a key aspect to take into account amongst possible environmental hazards to be encountered. Such particles can adhere to the lander, especially when moving from one test site to another, and start to build up on its surface. Such build-up of particles could end up leading to complications in the lander's mobility and system configuration. It must be mentioned the difficulty to mitigate such adhesion since lunar dust can be very abrasive and cohesive [6].

5.2.2. Hazard Mitigation:

The last Cassini fly-by of Enceladus occurred on Saturday, Dec. 19 at a distance of 3,106 miles (4,999 kilometers). This was the last of 22 Cassini fly-bys. In 2015, the Cassini probe made a close fly-by of Enceladus's south pole, flying within 48.3 km (30 mi) of the surface, as well as through a plume.

To protect the spacecraft against the radiation we will implement a radiation design margin (RDM) of two (minimum). This will ensure the protection of all electronic devices on the spacecraft. Total Ionizing Dose (TID) and Total Ionizing Dose Level (TIDL) must be considered when calculating RDM. RDM is the ratio of TID over TIDL. The use of an RDM is aimed at overcoming the inevitable uncertainties in environmental calculations and part radiation hardness determinations. TID is based on the maximum parameter degradation acceptable to ensure equipment operation in compliance with equipment performance specification at the end of the overall lifetime. TID is related to the design, instead of the component.

Testing will be conducted at the NASA Space Radiation Laboratory (NSRL), in Long Island, New York. This facility can simulate the harsh radiation found in outer space. Scientists use particle accelerators to make beams of protons and heavy ions. These are similar to the ones found in space radiation. The accelerator speeds them up to very high speeds. They are then sent through long tunnels to a target area. They strike the object being studied. The target area is shielded to keep radiation from escaping into the environment. Two materials will be tested to protect the spacecraft from radiation. One is aluminum. This metal is often used to build spacecraft because it is strong and lightweight. However, it can only absorb about half of the radiation that hits it. By testing other materials, scientists will be able to find new ways to make the spacecraft.

Each part of the spacecraft will go through radiation testing to ensure it can survive in space. After NSRL, we will conduct radiation testing at the Radiation Effects Facility, at NASA's Goddard Space Flight Center in Greenbelt, Maryland. We will use computer models to determine the amount of radiation we will encounter. For spacecraft, the main concern is particle radiation. This radiation, which includes protons and electrons, can impact their electronics in two ways. The first way is known as single event effects. These are immediate threats such as quick bursts of energy when a solar particle or cosmic ray pierces through a circuit. The second way is with effects that worsen with time. Charged particles can collect on a spacecraft's surface and build up a charge within hours. This is like walking across a carpeted room and turning a metal doorknob.

To combat the potential effects of radiation, parts that are hardened against radiation from their very foundations will be utilized. This means selecting materials that are less susceptible to particle strikes or charging. Shielding will also be implemented in the spacecraft design. Layering aluminum or titanium can slow down energetic particles, preventing them from reaching sensitive electronics. Shielding adds bulk and weight, which raises fuel needs or costs, so the engineers will calculate the necessary amount. Another strategy against single event effects is equipping an instrument with multiples of the same part that work together simultaneously. If one computer chip is temporarily disabled by a particle blow, its counterparts can pick up the slack.

Another consideration for the when trying to mitigate risks affecting the lander is the surface and strong winds encountered in Enceladus. Even though the surface is expected to be relatively smooth, we still need to prepare the rover to face obstacles and reduce the risk of the rover flipping over. To do so, a suspension system will be designed to help the lander account for the different surfaces it might encounter when moving from one test site to another. Regarding the strong winds, it was outlined that the lander would have a low center of mass and a small vertical surface area to account for this. Such design would aim to reduce the moment exerted on the lander by the wind and would again prevent it from rolling over.

It was already mentioned before the importance and difficulty of mitigating lunar dust adhesion from Enceladus's surface. This is a relevant topic since ongoing research is still taking place in order

to solve it. The team came up with an innovative idea which consists on coating the main parts of the lander with an Electrodynamic Dust Shield (EDS). Such device will consist of a set of conducting electrodes separated from one another by an insulating material [7]. As it is expected for the dust particles to contain some type of electrical charge, the EDS would create an electric field along the coating which would repel dust particles from adhering to the lander. From experiments done to the JSC Mars-1 simulants, by applying a voltage of 10kV corresponded to a 95% dust removal efficiency. Therefore, we can also expect similar results when trying to remove dust from Enceladus's surface.

Determining a minimum distance for the lander to be safe from the plumes was another task to be carried out by the team in order to mitigate possible environmental hazards that the lander might experience. As mentioned before, these plumes exert particles at incredible high speeds that can go hundreds of kilometers into space. This supposes an incredible force being generated at these plumes that could endanger the safety of the lander and its payload if placed too close to them. Smaller fissures can also be expected to surround these plumes, which would also exert water vapor into space. For these reasons, the team has considered it appropriate to have a safety distance limit of a 15km radius from the plumes. Since the particles being ejected from the plumes can reach hundreds of miles into space, it can be expected to find such particles in the surface from this safe distance. This distance will also guarantee the safety of the mission and would enable samples to be collected.

In what should be a similar construct as the Mars rover, there will be heaters packed to protect the probe's inner parts. A Warm Electronics Box will be made to protect (WEB) any batteries, electronics, and computers [8]. Along with this, excess heat from the electronics, and gold paint can be sources of providing warmth to the probe's main parts. A coating of gold paint significantly reduces the amount of energy radiated from the robot, therefore causing a form of insulation for the probe.

Section 6 - Activity Plan

6.1 - Budget

Personnel costs are estimated to be \$80,000 salaries per team member plus \$22,328 of employee related expenses (ERE) a year. With 3 people on the science team, 4 on engineering, and 4 on the administrative team this comes out to \$880,000 a year of expenses. The Project Manager and Deputy Project Manager are also on the science and engineering teams, respectively, but they are counted under the administrative for the purpose of cost estimation. Total yearly costs for personnel amounts to \$1,125,608.00.

Travel for this team is confined to traveling for the launch which involves each member getting a flight to Orlando, transportation to/from the airport to a hotel, and hotel expenses. Table 1 shows the detailed breakdown. There may be some necessary travel expenses outside of this for tests that require a team member to travel, but these additional expenses are unknown at this point. The cost for the entire team to be at the launch is at least \$11,292.50.

Other direct costs account for the majority of the expenses for this mission, with a total sum of just over \$51 million [1]. \$50 million of that goes towards the Lithium, with the rest being the science, thrusters, and other equipment costs [2]. None of these figures are thought to be final or even mostly correct at this point in time, as few pieces of equipment had similar instruments with easily accessible cost breakdowns. This means that a lot of estimation was used for the figures in the spreadsheet, including taking regular commercial prices for mass spectrometers and roughly doubling the cost of the most expensive ones to get a vague idea of the cost for what would fly on the lander. Values for facilities usage were based on very little actual information, and will be updated once we find a source for pricing.

The total cost of the project at this point in time is calculated to be \$160,475,814 [3], [4], [5]. This is including the default margins built into the provided spreadsheet. This is a good place for the team to start, as there is plenty of margin under the \$400 million budget, so if certain aspects of the estimation are inaccurate or the mission needs to be updated, there is financial overhead to do so. This also leaves enough extra for the 30% margin to be added on top and still be well within the budget.

Travel Costs

Flights

Table 5: Travel details and cost breakdown using 2020 date and times as a model.

Name	Flight (Roundtrip)	Cost	Total Running Cost	Arrival Time	Departure Time
Aina Zaidi	Detroit-Orlando	\$223	\$233	8:43 am	8:00 am
Peter Stokes	NY (JFK)-Orlando	\$171.50	\$394.50	9:56am	11:44 am
Abigail Roossien	Detroit-Orlando	\$223	\$617.50	8:43 am	8:00 am
German Paez Garcia	Columbus-Orlando	\$298	\$915.50	12:45 pm	11:50 am
Eduardo Vargas Gutierrez	NY (JFK)-Orlando	\$171.50	\$1087	9:56am	11:44 am
Luay Wadie	Michigan-Orlando	\$204	\$1291	1:05pm	7:00am
Mirelys Rosales	Maryland-Orlando	\$250	\$1541	9:17am	9:43 am
Kevin Ntoni	Boston-Orlando	\$207	\$1748	5:37 pm	10:50am
Yuri Kim	Boston-Orlando	\$207	\$1955	5:37 pm	10:50am
Mattison Boveri	Chicago-Orlando	\$204	\$2159	1:05pm	7:00am

Hotel

SpringHill Suites by Marriott Cape Canaveral Cocoa Beach

Rate: \$479/person for 4 nights

Transport

Uber SUV for 5 people (Aina, Abigail, Peter, Eduardo, Mirelys) to Hotel: \$80

Uber Sedan for 3 people (German, Luay, Mattison) to Hotel: \$57

Uber Sedan for 2 people (Kevin, Yuri) to Hotel: \$57

Uber Sedan for 3 people (Kevin, Yuri, Mirelys) to Orlando Airport: \$52

Uber Sedan for 3 people (Peter, Eduardo, German) to Orlando Airport: \$52

Uber Sedan for 4 people (Aina, Abigail, Luay, Mattison) to Orlando Airport: \$52

Per Diem

Rate: \$319.50/person for 5 days

6.2 Schedule

The mission to Enceladus is one that requires continued planning, research, and oversight. The process entailed is several months long and requires multiple phases. While some sections of the PDR will be requested earlier on, other sections will consist of an ongoing process. The deadline for every item up to and including the PDR and Video are under the L'SPACE program deadlines, but after that the mission deadlines become estimations of the desired cadence of the team.

Component	Start Date	End Date	Margin
Pre-Phase A: Conceptual Study	September 1st, 2020	December 3rd, 2020	
Phase A: Preliminary Analysis: Final Mission Concept, System-level Requirements, Project Management Plans	January 10th, 2021	April 1st, 2021	68 days
NASA: Review of Proposal	April 1st, 2021	June 6th, 2021	
Phase B: Preliminary Design	June 8th, 2021	September 18th, 2021	60 days
Specifications of Engineering and Instrument Requirements	June 8th, 2021	August 17th, 2021	
Preliminary Design Reviews for all Systems	July 14th, 2021	August 24th, 2021	
Phase C: Final Design and Fabrication	August 30th, 2021	December 18th, 2021	45 days
Critical Design Review	September 1st, 2021	October 5th, 2021	
Phase D: System Assembly, Integration and Test, Launch	February 2nd, 2022	August 20th, 2022	70 days
System Requirements System Design, Non-Advocate/Test Readiness Reviews	March 8th, 2022	April 8th, 2022	
Propulsion, Instruments, and Deployables Testing	April 11th, 2022	July 27th, 2022	
Launch	August 20th, 2022	August 20th, 2022	
Phase E: Operations Phase	September 30th, 2022	April 16th, 2023	167 days
Landing	April 16th, 2023	April 16th, 2023	
Data Analysis	April 20th, 2023	December 8th, 2023	
Mission Conclusion Report	December 10th, 2023	February 12th, 2023	
Outreach Materials, Press Conferences	September 14th, 2022	November 2nd, 2022	

Figure 12: Mission Deadlines

6.3 Outreach Summary

Throughout the duration of this project, there will be a four part outreach plan implemented. This plan includes social media outreach through Instagram and Youtube along with personal outreach in school programs.

For part one, an Instagram account will be made for the Oumuamua team. Through this account, the progress and milestones will be shared with the account's followers via Instagram stories and posts. The purpose of this part of the outreach plan is to increase interest in the project and target a younger audience of STEM learners. As of August 2020, 33.2 percent of Instagram users were between 25 and 34 years old. The second-largest user group had a 22.9 percent share and was between 18 to 24 years old. Another survey showed that 72 percent of teens age 13 to 17 used Instagram. This level of engagement and interaction with the application makes it an ideal location to promote and share the project [6]. By sharing daily videos, small interviews with the members of the team, and updates on the progress, the followers will become invested in the project and its success. Additionally, by sharing unique insider looks into the project from start to finish, people, especially the teen audience that is being targeted, will become more interested in STEM.

The second part of this plan includes Youtube videos. In addition to the daily video snippets on the Instagram account, a Youtube account will be made to add longer videos about certain aspects of the project and in order to capture and share long term progress. Creating and sharing these videos will also make them accessible to teachers in schools. Since it is not possible to visit every school in person, these videos will be a way that the project can enter every classroom across the U.S. The videos will show the behind the scenes of creating the spacecraft, from start to finish. This will spark interest and intrigue school-age students and the content will be available to everyone.

The last part of this outreach plan involves in-person school visits. Throughout the course of the project, team members will visit local schools and discuss the mission's progress and interact with students. They will perform small scale science experiments that relate to the project and share about Enceladus and why this mission is exciting. Interpersonal connections and hands-on experience will help promote STEM to young learners and excite the next generation of scientists and engineers.

Finally, we will engage with middle school, high school, and undergraduate students. Undergraduate students will be able to interact with the mission through modules and video conferences with the head scientists and engineers on the project. They will apply to be a part of a virtual fellowship community and will work together through the duration of the mission to work on parallel projects. This will engage the students in problem solving exercises similar to those being conducted for the Enceladus mission. This program will give undergraduate students the opportunity to get experience in the field of space exploration. After they complete the modules, there will be a Student Ambassador Virtual Community in order to connect current and former interns and fellows who have participated in the modules and contributed to the mission. Through this community they can network together and continue to work in the space exploration community.

Additionally, a year before the launch date, we will host a nationwide contest. This will involve middle and high school students. They will submit creative art pieces or essays that describe the importance of the Enceladus mission for the past, present, and future of our planet. This will help highlight the necessity of this mission and illustrate how it is different than any other mission before. Students should emphasize the unique characteristics of Enceladus and its environment that make it particularly interesting for scientists to learn about. The contest serves to engage students in space exploration and the current search for life elsewhere in the universe. The top three winners of the contest will be invited to the Enceladus launch.

6.4 Program Management Approach

Regarding the team's structural organization, it was decided to divide the team into three categories: Scientific, Engineering and Administrative. When assigning who would belong to each category, the preferences of each team member and their academic backgrounds were taken into account. Time availability was another factor to take into account as some members were not able to put the amount of time required for certain positions. Since there was a wide variety of STEM backgrounds, it was not too complicated to divide the team members into such sub-teams. When referring to the leadership positions, most of the members had no preferences when having to choose between taking a leader or follower position. Therefore making it easier to elect the Project Manager/ Deputy Project Manager positions and the Sub-Team Leads. The final outcome of the team's structural organization can be observed in Figure 2.

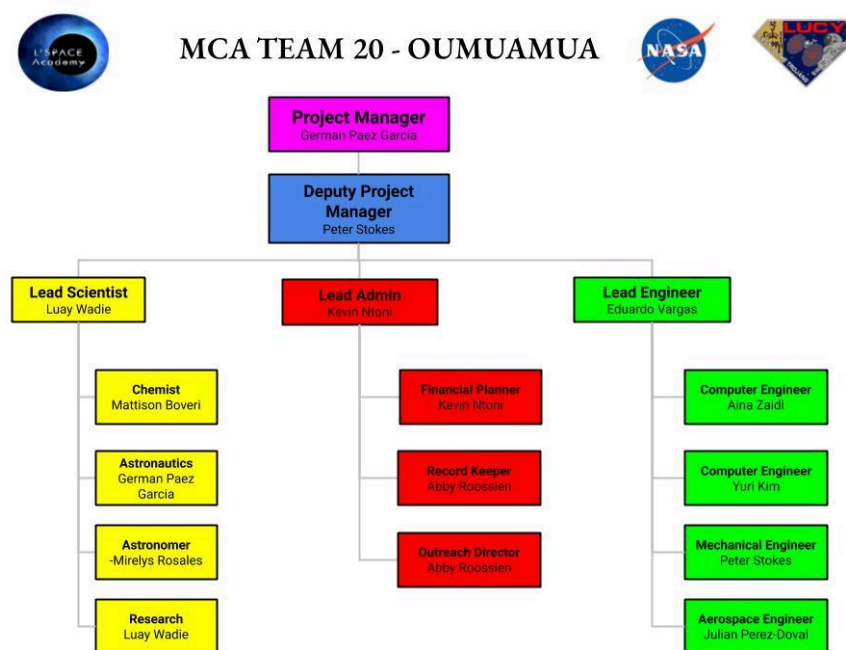


Figure 13:Organizational Chart

It must be mentioned that the team agreed that this organizational structure is not to be fixed. Meaning that if at some point a member from one category wanted to work with another sub-team, no objections would be given meanwhile this person's original sub-team was not left with excessive work to do. Finally, when referring to decisions made that will affect the whole team, it was agreed that the Project Manager and Deputy Project Manager will run the proposals through the rest of the team. The main reason for this is to get everyone to participate and be involved with the day to day activities of the team.

The team decided to make large decisions by having many individual proposals and taking a vote to decide what moves forward. Every team member designed a short concept proposal for the science payload of the lander and submitted it so the team could vote. The proposal with the most votes was used as the main guideline, and the proposal with the second most votes was used as inspiration wherever there was room to make modifications.

When distributing the workload, the main objective was to do so in a fair manner where everyone had to input similar amounts of effort. Even though each person will have a specific section to work

on, many of such segments will be related to each other. Therefore it was encouraged that team members would work with each other in order to complete the mandated assignments more effectively.

Finally, the topic of Conflict Resolution was covered by the team. The general consensus was to try and not involve any of the L'SPACE advisors or faculty in the team's internal conflicts. To do so, if there is a conflict between members, a third member will act as a mediator to work out the differences between the two parties. Such mediator was agreed to be one of the Project Managers or Sub-Team Leads. If any of the latter were to be involved in the conflict, the whole team would come together and mediate between the affected factions. In the case that the conflict is not resolved, the Project Manager would contact the team's advisor in order to try and find a solution. If this also fails to work, a faculty member from the L'SPACE Academy will be contacted to help resolve the situation.

7. Conclusion

Throughout this PDR, the team has addressed all the topics necessary to achieve a successful mission to Enceladus's Southern Pole. Here, the lander designed by the team will search for biosignatures and organics that could potentially survive in Enceladus's surface, enabling us to find any evidence of life. It was also determined that the age of the moon, its origins and the source from where heat originates. These clues would help us get a better understanding of Enceladus's environment and composition. To do this, the lander will visit five different locations where data will be collected and analyzed. Finally, the team decided to send the lander into one of the plumes located in the Baghdad Sulcus, where it will be obliterated but it will also be capable of recording more data and give us a better insight of these plumes.

This lander is of rectangular shape in order to maximize the utilization of permitted volume as part of the mission requirements. One thruster is placed in each corner of the lander in order to deorbit and land the craft on the surface of Enceladus. The entire craft is designed to be as simple as possible in order to maximize reliability and ease of manufacturing. The lander uses long landing legs with long crumple zones in order to absorb impact forces on touchdown. Once settled, the lander will deploy a drill out of the bottom of the hull using a telescoping scaffolding to bring the drill bit to the surface. Once a sample is obtained the drill will be retracted, the sample transferred to the GC-MS, and the GC-MS as well as the infrared spectrometer and the other cameras and sensors will begin collecting data.

Given more time, the team would like to develop and design the sample collection tool. The proposed method is a telescoping drill lowered to the surface and retracted once a sample is collected, and while this is a method employed by other craft, the exact procedure and designs for this craft are not definite at the moment. More time could allow for a specific and reliable sample collection which would ensure maximum success for the scientific payload.

Plans for the CDR revolve first around feedback given on the PDR. Any and all criticism would be weighed after review of the PDR, and all would be implemented into the CDR. This, as well as more iterating on current plans to refine the ideas would make for the best possible content for the CDR. After the content is at its highest achievable level, the presentation of the content must be prioritized. It is essential that the team is able to convey the goals and the processes this mission uses, so a lot of time would be spent editing for clarity. After edits to the document, the team would take time to practice presenting the material so each member can be confident and effective when talking about the mission.

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